

Experimental apparatus for universal spin-motion control in multi-ion registers



Donovan Webb
St Peter's College
University of Oxford

A thesis submitted for the degree of
Doctor of Philosophy in Physics

Trinity 2026

Contents

1	k-copy Randomised Benchmarking	1
1.1	Errors and Characterisation	2
1.2	Existing approaches to correlated-noise characterisation	5
1.3	The <i>k-copy</i> RB scheme	8
1.3.1	Protocol	8
1.3.2	Representations of $Cl_1^{\otimes 2}$	9
1.3.3	Common random unitary noise	12
1.3.4	Local depolarising noise	16
1.3.5	Combined model	16
	Asymptotes	18
	Schematic explanation	19
1.3.6	State preparation errors	21
1.4	Parity contrast: an independent probe of coherence	22
1.5	Sampling strategy	24
1.6	Scaling to $k > 2$ qubits	28
1.7	Experimental Results	31
1.7.1	Physical implementation	31
1.7.2	Z-basis measurements	32
1.7.3	Coherence measurements	34
1.7.4	Controlled noise injection	35
	Laser phase jitter	35
	Laser detuning scan	37
	Single addressed stark shifts	40
1.7.5	Three-qubit decay	41
1.7.6	Scaling to a multi-ion register	41
1.8	Conclusions and outlook	44

Appendices

A	Appendix	47
A.1	Generating Ions	47
A.2	Extracting Laser Offset and Magnetic Field	47
A.3	Experimental Control	47
A.4	Creating Squeezed States	47
A.5	Gate Fidelity	47
A.6	ZZ-type noise	50
	References	52

1

k-copy Randomised Benchmarking

Contents

1.1	Errors and Characterisation	2
1.2	Existing approaches to correlated-noise characterisation	5
1.3	The <i>k</i>-copy RB scheme	8
1.3.1	Protocol	8
1.3.2	Representations of $Cl_1^{\otimes 2}$	9
1.3.3	Common random unitary noise	12
1.3.4	Local depolarising noise	16
1.3.5	Combined model	16
1.3.6	State preparation errors	21
1.4	Parity contrast: an independent probe of coherence .	22
1.5	Sampling strategy	24
1.6	Scaling to $k > 2$ qubits	28
1.7	Experimental Results	31
1.7.1	Physical implementation	31
1.7.2	Z-basis measurements	32
1.7.3	Coherence measurements	34
1.7.4	Controlled noise injection	35
1.7.5	Three-qubit decay	41
1.7.6	Scaling to a multi-ion register	41
1.8	Conclusions and outlook	44

Check for missing citations between paper and chapter To construct useful scale quantum processors we must house many qubits physically close together, and must ensure the full register operates with “low error rates”. But how should we

define our errors in a large system? Naively, one might hope that the error rate of a multi-qubit system could be inferred from the error rates of its constituent qubits, but unfortunately, not all errors are commensurable — there are a zoo of possible error channels. They may be coherent, incoherent, continuous, stochastic, local, correlated, time dependent or independent (and I am sure a host of further adjectives exist). In response, a zoo of error correction and mitigation techniques have been developed each with their own scaling overheads and assumptions on these noise models they treat. It is a large and intimidating search space, and so, to better equip ourselves for navigation, we require tools to characterise what errors are relevant in our physical systems. In this chapter we introduce a new tool for characterising correlated errors in multi-qubit systems: *k-copy randomised benchmarking*. This technique extends single-qubit Clifford RB (discussed in chapter ??) to parallel application across multiple qubits, where now symmetry resolved dynamics provides evidence of, and quantifies various multi-qubit error channels.

This chapter is structured as follows: we start by giving an overview on the problem of *characterisation*, and what techniques already exist to address it. We then outline the *k-copy* protocol and discuss the relevant symmetric subspace structure. We derive the population dynamics when subject to simple noise models and discuss variations of the protocol that isolate specific subspaces. We provide experimental demonstrations of the protocol on a trapped-ion processor through injecting known correlated and uncorrelated errors. We end this chapter by demonstrating the scalability of the protocol on an 8-qubit register, and discuss the limitations and potential of this method for further scaling to larger systems.

1.1 Errors and Characterisation

Errors, which we shall give a broad definition as any deviations between the realised and ideal state of a quantum information processor (QIP), are due to unintentional environmental coupling and control imperfections. They are unavoidable, and if not carefully considered or mitigated, lead to unreliable or unverified computation. The process of modelling and quantifying these errors is known as *characterisation*.

Characterising errors has two practical advantages: 1) for the engineer, quantifying an error allows for a clear signal to improve hardware and calibration routines, and 2) for the theorist, understanding physically motivated error channels allows for the development of error correction and mitigation techniques that are robust to real world noise. Both users are concerned ultimately with the same question: How faithfully can we execute arbitrary circuits on our physical devices? Extracting metrics to this end is a wide research area in quantum information, with techniques of varying scalability and implicit assumptions on underlying noise models [1]. On one end of the spectrum sits full process tomography **citation** which reconstructs quantum channels, here represented as completely positive, trace preserving (CPTP) maps on the space of density matrices,

$$\mathcal{E} : \mathbb{C}^{d \times d} \rightarrow \mathbb{C}^{d \times d}, \quad (1.1)$$

where the dimension of the Hilbert space is $d = 2^n$ for n qubits. The CPTP map is rich in information, capturing many relevant error channels, but has poor scaling with system size, generally required $O(d^4)$ parameters to describe, and $O(d^4)$ measurements to reconstruct. This scaling fundamentally limits tomography to small system sizes. The goal then shifts to how to extract useful information in predicting the behaviour of a system executing a circuit, without full channel reconstruction. Now, two strategies emerge: 1) we make assumptions on the underlying noise model, or 2) we actively simplify the structure of the error channel through averaging. The first strategy leads to techniques such as noise learning **citation** and the latter leads to random circuit methods, such as randomised benchmarking (RB) [**magesan_characterizing_2012**, 2, 3]. Both approaches dramatically reduce the number of parameters required to describe the system, and the number of measurements required to extract them.

The assumptions on the noise model are cleanly motivated by considering the modularity of quantum circuits, and how errors may violate this modularity. Following the description of imperfect computation and crosstalk errors given in Refs. [4], and [5], respectively, we define the following models of QIPs, depicted

graphically in figure ??**add modularity figure**. Panel a) shows with the most desirable, yet technically unfeasible *Ideal QIP*, described by a *closed quantum system*. Here, the qubits comprise a quantum state represented by a single vector in Hilbert space. The state is isolated from the environment and may be decomposed into a collection of subsystems that remain uncoupled unless explicit multi-qubit operations are applied. Operations are all unitary evolutions, conveniently described by single and two-qubit unitaries, which we term *gates*. Errors may exist, but are limited to reversible unitary interactions. In b) we introduce external and control field noise due to an environment with no history, i.e. a source of Markovian noise. We must now depart from the quantum state vector and unitary evolution description, and instead use density matrices and CPTP maps to describe the state and dynamics of the QIP. We highlight a subset of these models, termed *Crosstalk-Free Markovian QIP*, where the dynamics of distinct subsystems are governed by locally acting CPTP maps. To satisfy this condition, operations are said to be *local* and *independent* [5].

Local is the assumption that only explicitly applied multi-body operations lead to correlations between qubits that participate in the ideal multi-body operation. For a bipartite system, a local Markovian QIP executing a circuit with no explicit coupling between subsystems, A , and B , can be described by the product channel CPTP map,

$$\mathcal{E}_{AB}(\rho_A \otimes \rho_B) = \mathcal{E}_A(\rho_A) \otimes \mathcal{E}_B(\rho_B), \quad (1.2)$$

where $\mathcal{E}_A$ and $\mathcal{E}_B$ are CPTP maps acting locally on the A and B subsystems respectively.

Independent is the assumption that the action of an applied gate on a subset of qubits is independent of any other gates applied in parallel on the other qubits. This implies that the CPTP map of gate G , acting on subsystem A in layer i , given by $\mathcal{E}_G^{(i)}(\rho_A)$, is identical to the action of the same gate acting on subsystem A in layer j ,

$$\mathcal{E}_G^{(i)}(\rho_A) = \mathcal{E}_G^{(j)}(\rho_A), \quad (1.3)$$

where parallel applied gates on the other subsystems may be different between layers.

With these two criteria satisfied, we may retain the composibility of gates, conventionally allowing us good prediction of the behaviour of a many-qubit deep circuit, using only characterisations of each gate in a universal gate set applied to small subsystems. If these criteria are not met, to accurately predict the behaviour of a deep circuit, each unique layer must be characterised, which is intractable.

The last class of models, included for completeness, are *Non-Markovian QIP*, where we relax the assumption that the effect of the environment has no history dependence. In this case, any instantaneous state may still be able to be described by a density matrix, however the dynamics can no longer be described by the application of a sequence of CPTP maps. To accurately characterise the system, full process tomography would need to be implemented for each possible circuit.

Practically, to ensure we can reasonably characterise our physical systems, and predict their performance on novel circuits, we must design and appropriately isolate hardware to be well approximated by the *Crosstalk-Free Markovian* models. The next section provides some background on existing techniques to verify crosstalk-free performance, and provide some quantification when the assumption is violated.

1.2 Existing approaches to correlated-noise characterisation

Existing approaches to correlated-noise characterisation span a trade-off between expressiveness and scalability. The most expressive, and arguably least scalable, are tomographic techniques, such as gate set tomography [nielsen_gate_2021] or the crosstalk sensitive variation [6]. Here, the implicit assumption that the noise environment is Markovian is sufficient, and no other assumptions of underlying noise models are required. These techniques are valuable in producing detailed CPTP maps that inform not only the presence of crosstalk-free violating noise, but also offer evidence on the underlying noise process. Demonstrations are however limited to small system sizes of two-qubits in [6].

Noise learning protocols are substantially more scalable by imposing a restricted

set of noise models, and fitting experimental data to these models. Correlated-noise specific variations of this approach [7, 8] apply repeated random circuits that act to simplify the action of noise channels to Pauli type errors, and then through measurements and statistical inference, likely Pauli error models are found, and crosstalk-free violating errors may be quantified. Noise learning can also be adaptable by first fitting simple models, for example when twirling over random circuits yields structured error channels, and progressively increasing the complexity of the inferred model when arbitrary circuits are run on the same device [9]. This nested-model approach provides a framework for robust characterisation of the system, however, demonstrations are still currently limited to small systems.

Similar to enforcing Pauli-type errors, randomised benchmarking techniques average the action of noise channels to impose structure to general error channels. This is achieved by twirling the action of the channel over deep circuits comprised of a chosen gate group. Clifford twirling techniques, notably the standard RB protocol [2, 10], have been adapted for detecting crosstalk induced correlations by simultaneous RB [11], and correlated RB [12] protocols. Correlated RB uses independent local twirling of multiple system copies to extract multi-qubit survival decay rates with Clifford sequence length. The weight and locality of correlated errors may then be inferred through a distance of the measured channel from a product (uncorrelated) channel. Other, non-Clifford twirled RB protocols, such as cycle-benchmarking [3] and EPLG [13] apply parallel gate sequences on multiple subsystems to extract some error applied per layer, that can then be used to infer error contributions due to correlated-noise channels. These methods have been shown to scale to 10s of qubits, and provide system level diagnostics of device performance.

We note two further streams of current research, first the application of RB in determining the unitarity of noise [14], and the extension of this work to quantifying bipartite unitary noise [[girling_estimation_2022](#)], which can induce correlations across the qubit register. Second, the use of custom circuits designed to probe crosstalk-free violations [[rudinger_probing_2019](#), 5, 15]. **expand section.**

The present protocol we have developed, and demonstrated experimentally, sits

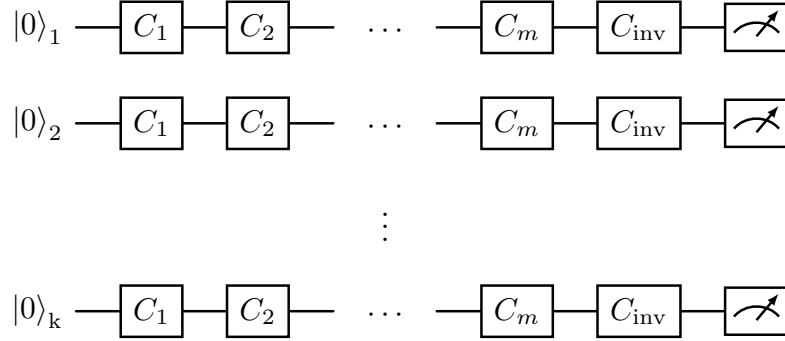


Figure 1.1: Circuit diagram for *k-copy* RB.

firmly in the RB methodology. It however differs from simultaneous and correlated RB by providing not only a quantification of crosstalk-free violating noise, but also providing evidence for the action of the error channel on a multi-qubit system. This extra information is inferred through direct observation of population decay and coupling between the subspaces enforced by twirling over the diagonally applied single-qubit Clifford group. The inspiration of this work comes from correlation spectroscopy [16] and subspace-leakage techniques [17, 18]. Correlation spectroscopy demonstrates that common-mode noise can be revealed through its action on collective and differential degrees of freedom, while subspace-leakage approaches provide a framework for defining error models not by full CPTP maps, but rather by their action on a discrete set of physically relevant subspaces. In such settings, symmetry is not merely a simplifying assumption providing structure to arbitrary channels, but rather it is the very mechanism that separates crosstalk-free violating errors from local noise. The next section details the subspaces enforced by diagonally applied single-qubit Clifford twirling, and provides the protocol for *k-copy* RB.

1.3 The *k-copy* RB scheme

This section starts with a circuit description of the *k-copy* RB protocol. We then develop the representation-theoretic framework required to analyze the decay curves arising from the correlated measurements. We finish this section by deriving the correlated populations observed experimentally when the system is subject to common random-unitary single-qubit, local depolarising, and entangling type noise.

1.3.1 Protocol

The *k-copy* RB protocol is implemented through standard single-qubit Clifford RB sequence (see section ??) applied simultaneously to multiple qubits, with the same randomly sampled Clifford applied to each qubit at every step, as shown in figure 1.1. A single sequence is composed of $m - 1$ parallel gates sampled randomly from the Clifford group. As in standard RB, the final applied layer is an inverting Clifford, which, due to the Clifford group being closed, may be selected to take the final state back to the initial computational eigenstate. The full sequence of random Clifford gates is then equivalent to the identity operation with some unknown noise overlaid¹. Practically, each gate layer can be executed using a global field, or through parallel local control lines addressing each qubit. A final $\langle Z \rangle$ measurement is made in parallel on all qubits to probe the successful execution of the Clifford sequence. Qubits measured in the target state are labelled $\mathcal{S}$, and qubits measured in the opposite state are labelled $\mathcal{F}$, for success and failure respectively. To build statistics of success and failure rates, r random circuits are trialled for each length m , and each circuit is repeated s times to reduce shot noise. To retain information on multi-qubit correlations, for every experimental shot, data from each qubit pair (i, j) is assigned to one of four outcomes, $\mathcal{P}_{SS}^{(i,j)}$, $\mathcal{P}_{SF}^{(i,j)}$, $\mathcal{P}_{FS}^{(i,j)}$, or $\mathcal{P}_{FF}^{(i,j)}$, where S and F denote success and failure. As with other RB protocols, the success probabilities versus sequence length are then fit to explicitly defined decay curves inferred by the action of the twirled error models.

¹Note: To further symmetrise state preparation and measurement errors, we may allow the final target state to be randomised between $|0\rangle$ and $|1\rangle$ for each sequence [10].

Move analysis of P_{SS} and $SE(P_{SS})$ here.

By limiting the analysis to pairwise comparisons, the *k-copy* protocol is both efficient in the number of trials required on the device, and in the postprocessing of the recovered data. For a register of k qubits, the same shot record contains outcomes for all $k(k-1)/2$ pairs. The two-body error models are derived in the following sections for prevalent noise mechanisms in both ion trapping and superconducting information processors. Higher weight correlations are discussed further in section 1.6.

1.3.2 Representations of $Cl_1^{\otimes 2}$

Here, we discuss the representations of the single-qubit Clifford group acting diagonally on two copies, $Cl_1^{\otimes 2}$, first presented in reference [19]. The irreducible representations give a natural (Schur) basis to consider how possible error channels are simplified by *k-copy* Clifford twirling.

We consider a group $\mathcal{G}$ and a representation $r : \mathcal{G} \rightarrow GL(\mathcal{L})$, where GL is the general linear group (consisting of the set of invertible matrices and the matrix multiplication operation), and $\mathcal{L}$ denotes the Liouville (vectorised operator) space of two qubits. Twirling over the elements g in $\mathcal{G}$ enforces symmetry on a general error channel, Λ , given in Liouville form,

$$\tilde{\Lambda} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} r(g) \Lambda r(g)^\dagger, \quad (1.4)$$

where $|\mathcal{G}|$ is the number of elements comprising the finite group $\mathcal{G}$. The twirling enforces commutation with the group action,

$$r(g) \tilde{\Lambda} r(g)^\dagger = \tilde{\Lambda} \quad \forall g \in \mathcal{G}. \quad (1.5)$$

The representation of the space, $\mathcal{L}$ decomposes to irreps λ as,

$$\mathcal{L} = \bigoplus_{\lambda} (V_{\lambda} \otimes \mathbb{C}^{m_{\lambda}}), \quad (1.6)$$

where V_{λ} is the irreducible representation space, and $\mathbb{C}^{m_{\lambda}}$ is a complex valued multiplicity space with dimension of the multiplicity m_{λ} . From Schur's lemma,

Sub-representation	irrep	Basis operators
Identity	1	II
Left qubit	3	XI, YI, ZI
Right qubit	3	IX, IY, IZ
Trace	1	$(XX + YY + ZZ)/\sqrt{3}$
Diagonal traceless	2	$(XX + YY - 2ZZ)/\sqrt{6},$ $(XX - YY)/\sqrt{2}$
Symmetric off-diagonal	3'	$(XY + YX)/\sqrt{2},$ $(XZ + ZX)/\sqrt{2},$ $(YZ + ZY)/\sqrt{2}$
Antisymmetric	3	$(XY - YX)/\sqrt{2},$ $(XZ - ZX)/\sqrt{2},$ $(YZ - ZY)/\sqrt{2}$

Table 1.1: Sub-representations of the $Cl_1^{\otimes 2}$ group. The irrep, dimension of each sub-representation, and a set of operators in the Pauli basis which span the subspace dimension is given. The irrep basis vectors are from equation 56 in reference [20].

Class	Representation	Member
Identity	e	I
Swap	(12)	H
Two-swap	(12)(34)	X
Three-cycle	(123)	HS
Four-cycle	(1234)	S

Table 1.2: Conjugacy classes for the single qubit Clifford group.

equation 1.5 implies the twirled error channel has the form,

$$\tilde{\Lambda} = \bigoplus_{\lambda} (I_{\lambda} \otimes \mathcal{A}_{\lambda}), \quad (1.7)$$

where $\mathcal{A}_{\lambda}$ is the multiplicity matrix of dimensions $m_{\lambda} \times m_{\lambda}$.

irrep	e	(12)	(12)(34)	(123)	(1234)
Identity/Trace	1	1	1	1	1
Left/Right/Antisymmetric	3	-1	-1	0	1
Diagonal traceless	2	0	2	-1	0
Symmetric off-diagonal	3	1	-1	0	-1

Table 1.3: Character table for the Clifford group given the irreps in table 1.1, and with conjugacy classes: identity, transposition, double transposition, 3-cycle, and 4-cycle. The “sign” irrep of the single-qubit Clifford group has zero multiplicity in the two-copy case, and so is excluded here.

The 2-copy single-qubit Clifford group, analysed in detail in references [19, 20], decomposes the general 16-dimensional two qubit space into 4 irreducible representations, irreps, with dimensions $\{1, 2, 3, 3'\}$, (the notation $3'$ is used to contrast the other 3-dimensional irrep), and multiplicities $\{2, 1, 3, 1\}$. The subrepresentations are summarised in table 1.1. **Could Move character table and conjugacy classes to appendix** The irreps are uniquely determined by their character on the conjugacy classes of the Clifford group. Two group elements $g, h \in G$ are said to be in the same conjugacy class if they are related by conjugation,

$$g = khk^{-1}, \quad (1.8)$$

where $k \in G$. With this definition, the 24 single qubit Clifford elements are divided into five permutation classes, shown in table 1.2, with a representative name and example member following each class. These classes make obvious the known isomorphism between Cl_1 and the symmetric group S_4 (modulo phase). Character tables [lee_character_2023], are useful tools in representation theory, to uniquely describe irreps by the traces of the representative matrices for group elements of each conjugacy class. The table is presented as a 2D matrix with rows corresponding to group representations and columns to group conjugacy classes. The character, ξ , is defined by,

$$\xi_\lambda(g) = \text{Tr}[r_\lambda(g)], \quad (1.9)$$

where r_λ is the irreducible component of the representation of G acting on some vector space,

$$r_\lambda(g)_{i,j} = \text{Tr}[B_i^\dagger g B_j g^\dagger], \quad (1.10)$$

where B_i are the orthonormal basis vectors for irrep λ . Taking the two-qubit subrepresentations reproduced in table 1.1, the character table calculated is shown in table 1.3. We can see confirmed that the left, right, and antisymmetric subrepresentations form a multiplicity of the 3D-irrep labelled 3, while the trace subrepresentation is a multiplicity of the 1D trivial irrep.

Given the structure of a general twirled map is enforced to be block diagonal in the irrep basis, given by equation 1.7, we can do a simple count to find generally how many parameters are required to fully describe the CPTP map. Accounting for the multiplicities, and the constraint that the map is trace preserving, we find that at most $2^2 + 3^2 + 1 + 1 - 2 = 13$ independent scalars are needed to describe $\tilde{\Lambda}$. The diagonal terms, here labelled $\vec{\alpha} = (\alpha_L, \alpha_R, \alpha_{tr}, \alpha_{dt}, \alpha_{so}, \alpha_{as})$, describe a decay rate contribution from each sub-representation, whilst off-diagonal terms, $\vec{m} = (m_{id \rightarrow tr}, m_{L \rightarrow R}, m_{R \rightarrow L}, m_{L \rightarrow as}, m_{R \rightarrow as}, m_{as \rightarrow L}, m_{as \rightarrow R})$ describe coupling between the multiplicity spaces.

In practice, different physical error mechanisms populate only a subset of the coefficients α and m , leading to simplified dynamics with characteristic signatures of the underlying process. Table 1.4 provides possible physical mechanisms capable of coupling isotypic representation subspaces. We devote some time in the next two sections to describing noise due to fluctuations of a common field, and to local depolarising type errors. In appendix A.6 we discuss a ZZ -type noise model, relevant to the superconducting qubit platform.

1.3.3 Common random unitary noise

It can be seen that the error channels that lead to couplings neatly discern the physical origin of the noise channel. For example, in transmon superconducting qubit platforms, always on ZZ -couplings are a dominant crosstalk error source [fors_comprehensive_2024, alghadeer_crosstalk_2025, 21], and so the couplings between single qubit subspaces and the antisymmetric subspace can be quantified; in the neutral atom platform, correlated non-unital errors may be observed if the physical qubit separation is of similar order to the qubit transition wavelength [22]; however, in ion trap platforms, environmental noise is often dominated by magnetic and electric field fluctuations, which may act globally on all qubits if they are stored in the same trapping potential. These field fluctuations induce both common and independent single-qubit rotations, but do not couple irrep subspaces together (i.e $m = 0$ for $m \in \vec{m}$). Thus, the general error

Coupling	irrep	Mechanism
$m_{id \rightarrow tr}$	1	Decays (non-unital)
$m_{L \rightarrow R},$ $m_{R \rightarrow L}$	3	Qubit Swap, Correlated decays (non-unital)
$m_{L \rightarrow as},$ $m_{R \rightarrow as},$ $m_{as \rightarrow L},$ $m_{as \rightarrow R}$	3	Entanglement

Table 1.4: Possible mechanisms that result in the coupling of sub-representation sectors of the $Cl_1^{\otimes 2}$ twirled error channel. **Add evidence to support this claim?**

map of this type is now parametrised by at most the 6 scalars, $\vec{\alpha}$, and results in Liouville form diagonal in the Schur basis, $\mathcal{S}$, (ordered by table 1.1),

$$\tilde{\Lambda}^{\mathcal{S}} = \text{diag} (I_1, \alpha_L I_3, \alpha_R I_3, \alpha_{tr} I_1, \alpha_{dt} I_2, \alpha_{so} I_3, \alpha_{as} I_3), \quad (1.11)$$

where I_d is the d -dimensional identity. Specifically, we consider the effect of noise due to common small-angle single-qubit rotations acting on two-qubits after each gate comprising the desired circuit. The general form for this error channel is given by the separable channel,

$$\mathcal{E}_{U \otimes U}^{\mathcal{A}}(\rho) = \sum_{a \in \mathcal{A}} p_a (U_a \otimes U_a) \rho (U_a \otimes U_a)^\dagger, \quad (1.12)$$

where $\mathcal{A}$ specifies the ensemble of unitaries applied with probabilities p_a , with $\sum_{a \in \mathcal{A}} p_a = 1$, and $U_a(\theta_a, \hat{n}_a) = e^{-i(\theta_a/2)\hat{n}_a \cdot \hat{\sigma}}$, with rotation angle, θ_a , axis, $\hat{n}_a$, and Pauli vector, $\hat{\sigma}$. A subset of error channels following this description cannot be described by a product channel map on a bipartite system, and so violate the *local* condition given in equation 1.2. These errors therefore cannot be modelled by a crosstalk-free Markovian QIP although originating from separable channels. This is an important distinction: non-product channel are not limited to direct entangling noise.

We now consider the effect of twirling on a single common unitary to build a structured error channel. The single qubit rotation is parametrised by rotation angle, θ , and the axis of rotation $\vec{n}$,

$$U(\theta, \hat{n}) = e^{-i\frac{\theta}{2}\hat{n} \cdot \hat{\sigma}}. \quad (1.13)$$

The channel comprised of unitary rotations U_a ,

$$\mathcal{E}_a(\rho) = (U_a \otimes U_a)\rho(U_a \otimes U_a)^\dagger, \quad (1.14)$$

is twirled over the discrete group $\mathcal{G}$, with twirling defined by equation 1.4, leading to the simplified channel in Liouville form $\tilde{\Lambda}_a$. Reintroducing the stochastic application of common unitaries, given by equation 1.12, we find,

$$\tilde{\Lambda} = \sum_{a \in \mathcal{A}} p_a \tilde{\Lambda}_a. \quad (1.15)$$

Rewrite section in own words. Could move to appendix? Thus, it is sufficient to consider the decay due to a single rotation acting identically on both qubits which, in the Pauli-transfer matrix (PTM) representation, can be written as $\Lambda^{\mathcal{P}} = \text{diag}(1, R, R, R \otimes R)$, where superscript $\mathcal{P}$ represents the block diagonal matrix is ordered by the two-qubit Pauli basis $\mathcal{P} = \{I \otimes I, \vec{\sigma} \otimes I, I \otimes \vec{\sigma}, \vec{\sigma} \otimes \vec{\sigma}\}$, with $\vec{\sigma} = \{\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z\}$, and $R = R(\vec{n}, \theta) \in SO(3)$ describes a rotation by θ around $\vec{n}$. Finding the vector $\vec{\alpha}$ can now be done using Schur's lemma for the case of no mixing between irrep spaces:

$$\tilde{\Lambda}^{\mathcal{P}} = \sum_{\lambda} \alpha_{\lambda} \Pi_{\lambda}, \quad \alpha_{\lambda} = \frac{1}{d_{\lambda}} \text{Tr}(\Pi_{\lambda} \Lambda^{\mathcal{P}}). \quad (1.16)$$

Here, Π_{λ} denotes the PTM projector onto the irrep space labelled λ and with dimension d_{λ} . It follows immediately that the identity sector is fixed, while the left- and right-qubit sectors produce $\alpha_L = \alpha_R = \text{Tr}(R)/3$. In principle, α_{λ} for the two-qubit sector can be evaluated by explicitly constructing the projectors from the basis operators in Table 1.1. However, in this case the symmetries of these irrep spaces allow the projectors to be expressed more simply in terms of operators whose traces are easily evaluated:

$$\Pi_{tr} = T, \quad (1.17)$$

$$\Pi_{dt} = D - T, \quad (1.18)$$

$$\Pi_{so} = \frac{1}{2}(I_9 + S) - D, \quad (1.19)$$

$$\Pi_{as} = \frac{1}{2}(I_9 - S), \quad (1.20)$$

where $T = 1/3 \sum_{ij} |ii\rangle \langle jj|$, $D = \sum_i |ii\rangle \langle ii|$, and $S = \sum_{ij} |ji\rangle \langle ij|$ are the trace projector, diagonal projector, and swap operator, respectively. The required traces are

$$\text{Tr}(I_3 R) = \text{Tr}(R) = (1 + 2c), \quad (1.21)$$

$$\text{Tr}(I_9(R \otimes R)) = (\text{Tr}(R))^2 = (1 + 2c)^2, \quad (1.22)$$

$$\text{Tr}(T(R \otimes R)) = \frac{1}{3} \text{Tr}(R^T R) = 1, \quad (1.23)$$

$$\text{Tr}(D(R \otimes R)) = \sum_i R_{ii}^2 = c^2 + 2c + (1 - c)^2 \kappa, \quad (1.24)$$

$$\text{Tr}(S(R \otimes R)) = \text{Tr}(R^2) = 4c^2 - 1, \quad (1.25)$$

where $c = \cos(\theta)$, and $\kappa = n_x^4 + n_y^4 + n_z^4$ is an anisotropy parameter reflecting the fact that Clifford twirling enforces symmetry under permutations of the Pauli axes, rather than under arbitrary rotations. Substituting these traces into equation 1.16 and using equation 1.17 yields

$$\alpha_L = \alpha_R = \alpha_{as} = \frac{1}{3}(1 + 2c), \quad (1.26)$$

$$\alpha_{tr} = 1, \quad (1.27)$$

$$\alpha_{dt} = \frac{1}{2}(c^2 + 2c - 1 + (1 - c)^2 \kappa), \quad (1.28)$$

$$\alpha_{so} = \frac{1}{3}(3c^2 - (1 - c)^2 \kappa). \quad (1.29)$$

Taking these scalar products to second order in error angle θ , we find that the anisotropic term may be ignored for small errors,

$$\alpha_L = \alpha_R = \alpha_{as} = 1 - \theta^2/3, \quad (1.30)$$

$$\alpha_{tr} = 1, \quad (1.31)$$

$$\alpha_{dt} = \alpha_{so} = 1 - \theta^2. \quad (1.32)$$

As such, the dynamics are entirely described by the error angle magnitude. The ensemble of twirled common unitaries across some arbitrary distribution, $\mathcal{A}$, given in equation 1.12, thus combine to give scalar parameters describing the contribution from each subspace to the observed decay,

$$\alpha_i^{(\mathcal{A})} = \sum_{a \in \mathcal{A}} p_a \alpha_a. \quad (1.33)$$

This definition leads to a convenient (and conforming to previous RB work) definition for the correlated error per Clifford given by,

$$\varepsilon_{\text{corr}} = \frac{1}{6} \sum_{a \in \mathcal{A}} p_a \theta_a^2, \quad (1.34)$$

where the correlated error per Clifford, $\varepsilon_{\text{corr}}$, quantifies the single-qubit error contribution due to *common noise*. Comparing to equation 1.11, the general ensemble of common single-qubit rotations channel, after twirling over $Cl_1^{\otimes 2}$, is given by the diagonal matrix in the Schur basis,

$$\begin{aligned} \tilde{\Lambda}_{U \otimes U}^S = \text{diag} \Big(& I_1, (1 - 2\varepsilon_{\text{corr}})I_6, I_1, \\ & (1 - 6\varepsilon_{\text{corr}})I_2, (1 - 2\varepsilon_{\text{corr}})I_6 \Big). \end{aligned} \quad (1.35)$$

1.3.4 Local depolarising noise

Ultimately, environmental noise will not perfectly couple symmetrically to both qubits, and so independent local single-qubit error channels must also be considered. Assuming an independent noise channel acting on each qubit, Clifford twirling leads to a fully depolarising noise channel of the form,

$$\mathcal{E}_{dp,a}(\rho^{(1q)}) = (1 - 2\varepsilon_a)\rho^{(1q)} + \varepsilon_a I_2 \text{Tr}(\rho^{(1q)}), \quad (1.36)$$

where the superscript denotes that this is for a single qubit subspace, and the subscript a signifies which qubit we are considering. For the “left” and “right” qubits the decay rates are parametrised by ε_L and ε_R , respectively. Comparing to equation 1.11, the independent depolarisation channel may be expressed in Liouville form as the diagonal matrix in the Pauli basis,

$$\Lambda_{dp}^P = \text{diag} \Big(1, (1 - 2\varepsilon_L)I_3, (1 - 2\varepsilon_R)I_3, (1 - 2\varepsilon_L)(1 - 2\varepsilon_R)I_9 \Big). \quad (1.37)$$

1.3.5 Combined model

Combining the depolarisation channels with the twirled common single-qubit rotation channel,

$$\Lambda_T(\varepsilon_L, \varepsilon_R, \varepsilon_{\text{corr}}) = \Lambda_{dp}(\varepsilon_L, \varepsilon_R) \tilde{\Lambda}_{\text{corr}}(\varepsilon_{\text{corr}}), \quad (1.38)$$

we find scalar parameters,

$$\alpha_L = 1 - 2(\varepsilon_L + \varepsilon_{corr}), \quad (1.39)$$

$$\alpha_R = 1 - 2(\varepsilon_R + \varepsilon_{corr}), \quad (1.40)$$

$$\alpha_{tr} = 1 - 2(\varepsilon_L + \varepsilon_R), \quad (1.41)$$

$$\alpha_{dt} = \alpha_{so} = 1 - 2(\varepsilon_L + \varepsilon_R + 3\varepsilon_{corr}), \quad (1.42)$$

$$\alpha_{as} = 1 - 2(\varepsilon_L + \varepsilon_R + \varepsilon_{corr}). \quad (1.43)$$

To reproduce the decay dynamics observed in the RB sequence, we must consider an initial prepared state, and the final measurement basis. For the RB sequence given in figure 1.1, the initial state is prepared as $|00\rangle$, and the final correlated populations are measured in the computational eigenstates, $\{|00\rangle \leftrightarrow SS, |01\rangle \leftrightarrow SF, |10\rangle \leftrightarrow FS, |11\rangle \leftrightarrow FF\}$, where S and F denote “success” and “failure” of recovering the target state outcome for each qubit. The correlated populations may be given analytically using the derived Liouville operators, and by the vectorized initial and final states,

$$\mathcal{P}_{LR} = \langle\langle L, R | \Lambda_T^m | 00 \rangle\rangle, \quad (1.44)$$

where L, R denote the successful recovery of the target final state for the “left” and “right” qubit respectively, and m is the number of Clifford gates in the applied sequence. Using this equation, and the explicit transfer matrices, the final populations decays are found,

$$\mathcal{P}_{SS} = \frac{1}{4} + \frac{1}{4}(\alpha_L^m + \alpha_R^m) + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m, \quad (1.45)$$

$$\mathcal{P}_{SF} = \frac{1}{4} + \frac{1}{4}(\alpha_L^m - \alpha_R^m) - \frac{1}{6}\alpha_{dt}^m - \frac{1}{12}\alpha_{tr}^m, \quad (1.46)$$

$$\mathcal{P}_{FS} = \frac{1}{4} - \frac{1}{4}(\alpha_L^m - \alpha_R^m) - \frac{1}{6}\alpha_{dt}^m - \frac{1}{12}\alpha_{tr}^m, \quad (1.47)$$

$$\mathcal{P}_{FF} = \frac{1}{4} - \frac{1}{4}(\alpha_L^m + \alpha_R^m) + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m. \quad (1.48)$$

Considering the final state of the single qubit subspace, as probed by standard RB, we find the expected single exponential decay,

$$\mathcal{P}_S^{(L)} = \mathcal{P}_{SS} + \mathcal{P}_{SF} = \frac{1}{2}(1 + (1 - 2(\varepsilon_L + \varepsilon_{corr}))^m), \quad (1.49)$$

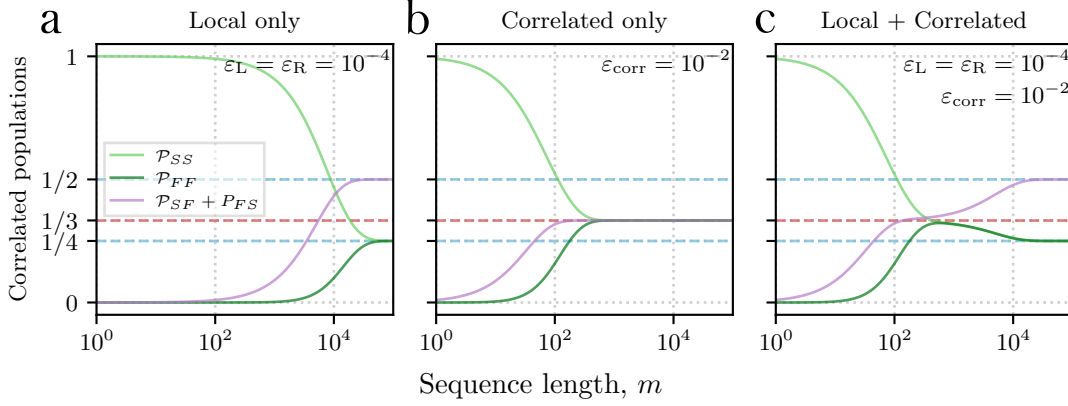


Figure 1.2: Numerical simulation of intermediate asymptotes measured in Z basis.

where the superscript (L) refers to considering the left qubit only. Comparing to standard single qubit RB parameterisation,

$$\mathcal{P}_S = \frac{1}{2} (1 + (1 - 2\varepsilon_c)^m), \quad (1.50)$$

where ε_c is the “error-per-Clifford”, we find the observed decay of each qubit is now separable into a local, and correlated error component,

$$\varepsilon_c^{(L)} = \varepsilon_L + \varepsilon_{corr}. \quad (1.51)$$

Asymptotes

Here we consider the asymptotic structure of the correlated decay curves given in 1.45. Under dominant local depolarisation, the two-qubit state will approach the maximally mixed state, indicated by the asymptote $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/4$, as seen in figure 1.2 (a). Under dominant *common noise*, with decay profile seen in (b), an asymptote of $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/3$ is approached due to the invariance of the symmetric trace sector, α_{tr} . Panel (c) shows the case where both local depolarising and *common noise* is present. When $\varepsilon_{corr} > \varepsilon_L = \varepsilon_R$ a clear intermediate asymptote is seen before a final decay to the fully depolarised state. This intermediate asymptote is the central experimental signature of correlated noise exploited throughout this chapter, and is explored experimentally in section 1.7.2 below.

The mixed state reached after the application of *common noise* to the initial state $|00\rangle$ can be understood by directly inspecting the space of possible pure two qubit states reached through common only rotations. The general pure state after common rotations is the separable identical two-qubit states,

$$|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle, \quad (1.52)$$

with $|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$ for a single qubit general state. Depolarising into the symmetric separable two-qubit states is equivalent to averaging over the full Bloch sphere,

$$\begin{aligned} \rho_{\text{sym}} &= \frac{1}{4\pi} \int_0^\pi \int_0^{2\pi} (\sin(\theta) |\Psi\rangle \langle\Psi|) d\phi d\theta \\ &= \frac{1}{3} |00\rangle \langle 00| + \frac{1}{3} |11\rangle \langle 11| + \frac{1}{3} |\Psi^+\rangle \langle\Psi^+|, \end{aligned} \quad (1.53)$$

where the third term is the Bell state $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$. The presence of this Bell-state component reflects residual coherence induced between the two qubits by the shared rotation error, and motivates the alternative strategy of a coherence-sensitive probe developed in section 1.4.

Schematic explanation

The distinction between correlated and uncorrelated noise is illustrated schematically in figure 1.3. We consider a particularly simple unitary noise mechanism to highlight the qualitative difference of information gained between standard and *k-copy* RB. After each Clifford gate we consider an independent unitary rotation errors on each qubit, and a common unitary rotation error on qubits $i = 3$ and $i = 4$. Each unitary rotation axis is randomly drawn (isotropically from $SO(3)$), and the angle of the rotation is small. The error channels are explicitly included in the circuit diagram in (a). The state of the qubits after the twirling procedure, and before measurement, are shown in (b). We see that all qubits are no longer in the eigenstate $|0\rangle$ due to the unitary noise **add bloch sphere to fig**. We define the angular deviation of qubit i from $|0\rangle$ as θ_i **add theta to fig**. The insets in (b) highlight a qualitative difference between common and independent noise — the joint angle distribution

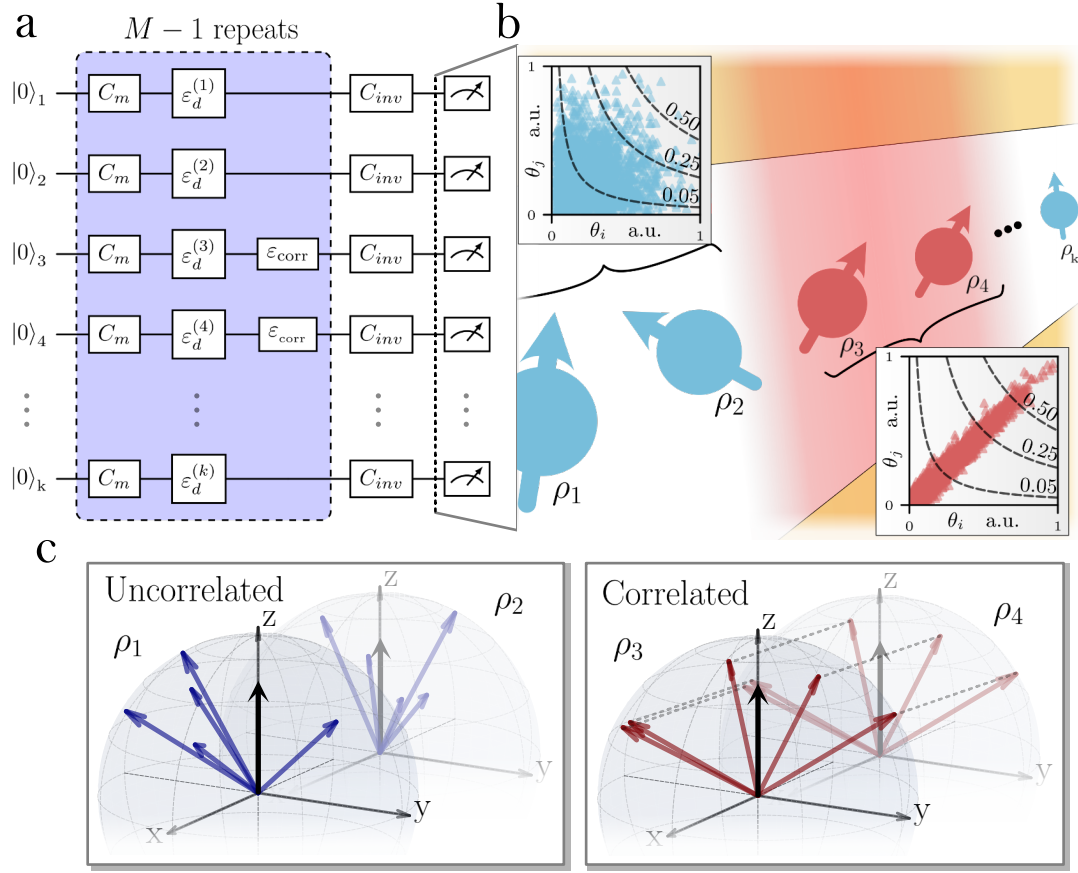


Figure 1.3: Bloch sphere schematic of corr vs uncorr noise. Include schematic of ion chain prev fig 1c. Change corr to unitary maybe?

of a qubit pair after many repeated trials either appears circular in the case of independence, or appear highly elliptical if common unitary noise dominates. The dashed lines indicate the probability for two-qubit errors, which are more prevalent in the case of common mode noise **quantify more carefully**.

Panels (c) and (d) display the final vectors resulting from 5 random sequences. All qubits here exhibit identical single-qubit RB error rates, which are probed by standard RB, and shown here through the identical depolarised Bloch vectors (black arrows). However, looking at pairwise correlations, we see: independent noise produces uncorrelated sequence-to-sequence deviations, whereas common noise produces shared deviations between qubits acted on by the same random Clifford sequence. The parallel twirling during *k*-copy RB preserves exactly this

symmetry information that would be averaged away under independently sampled local Clifford sequences.

1.3.6 State preparation errors

Similar to previous work on subspace leakage benchmarking [17], state preparation errors can lead to non-trivial modifications of the correlated population decay curves. First, we consider independent state preparation on two qubits. We model independent errors as probabilistic bit flips. To first order in error, on a single qubit we have,

$$\rho_{sp}^{(i)} = (1 - \varepsilon_{sp}^{(i)}) |0\rangle \langle 0| + \varepsilon_{sp}^{(i)} |1\rangle \langle 1|, \quad (1.54)$$

where the superscript refers to which qubit, L , or R , the preparation error is relevant to. This error may be modelled by $\mathcal{E}_{sp}^{(i)}$, a probabilistic X channel applied to an arbitrary state,

$$\mathcal{E}_{sp}^{(i)}(\rho) = (1 - \varepsilon_{sp}^{(i)}) \rho + \varepsilon_{sp}^{(i)} X \rho X. \quad (1.55)$$

Transforming to the Liouville form, denoted by $\Lambda_{sp}^{(i)}$, this error is simply incorporated into the population dynamics given by equation 1.44, by acting on the initial state,

$$\mathcal{P}_{LR} = \langle \langle L, R | \Lambda_T^m \Lambda_{sp}^{(L)} \Lambda_{sp}^{(R)} | 00 \rangle \rangle. \quad (1.56)$$

In this picture, the state preparation errors scale only the overlap of the initial state with the sub-representations. The overlaps are multiplied by,

$$A_{sp}^{(i)} = (1 - 2\varepsilon_{sp}^{(i)}), \quad (1.57)$$

if the qubit is involved non-trivially in the subspace projector, i.e. for the left and right qubit sectors $A_{sp} = A_{sp}^{(i)}$. Relevant to this work, the correlated survival populations with state preparation errors included are,

$$\begin{aligned} \mathcal{P}_{SS} = & \frac{1}{4} + \frac{A_{sp}^{(L)}}{4} \alpha_L^m + \frac{A_{sp}^{(R)}}{4} \alpha_R^m \\ & + \frac{A_{sp}^{(L)} A_{sp}^{(R)}}{6} \alpha_{dt}^m + \frac{A_{sp}^{(L)} A_{sp}^{(R)}}{12} \alpha_{tr}^m. \end{aligned} \quad (1.58)$$

Similar forms exist for populations, $\mathcal{P}_{SF}$, $\mathcal{P}_{FS}$, and $\mathcal{P}_{FF}$. Significantly, this state preparation error does not add projection on the antisymmetric or symmetric off-diagonal sectors, which would introduce another exponential component into the observed decay curves.

Correlated state preparation errors may be considered by adapting equation 1.55 to a two-qubit stochastic error channel, such as correlated spin-flips given by

$$\mathcal{E}_{sp}^{corr}(\rho) = (1 - \varepsilon_{sp}^{corr})\rho + \varepsilon_{sp}^{corr}(X \otimes X)\rho(X \otimes X). \quad (1.59)$$

The subspaces involving both qubits non-trivially are now unaffected by the state preparation error, shown explicitly by,

$$\mathcal{P}_{SS} = \frac{1}{4} + \frac{A_{sp}^{corr}}{4}\alpha_L^m + \frac{A_{sp}^{corr}}{4}\alpha_R^m + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m, \quad (1.60)$$

where $A_{sp}^{corr} = 1 - 2\varepsilon_{sp}^{corr}$. These correlations thus appear similar to *common noise* errors in the invariance of α_{tr} to this channel. Therefore correlated state prep errors can lead to a systematic bias in quantification of correlated error per Clifford if not accounted for in the fit by an initial state preparation error offset.

1.4 Parity contrast: an independent probe of coherence

Population decays described by equation 1.45 indicate an intermediate asymptote corresponding to the suggested state, ρ_{sym} , given by equation 1.53. Here we propose a parity-based measurement to independently verify the coherent structure seen in ρ_{sym} . Specifically, a reduced tomography method based on [23] is used to quantify overlap of the final measured state with the Bell state, $|\Psi^+\rangle$, component in ρ_{sym} .

The circuit for probing this coherence is shown in figure 1.4 (a): differential rotations of the form $\hat{R}_{+\phi}(\pi/2) \otimes \hat{R}_{-\phi}(\pi/2)$, with rotation axis $\hat{\sigma}_\phi = \cos(\phi)\hat{\sigma}_x + \sin(\phi)\hat{\sigma}_y$, are applied prior to measurement, and the contrast of the resulting fringe (as ϕ is varied) is extracted.

Figure 1.4(b) show the expected contrast of parity measurements as a function of Clifford sequence length. The analytic model of the parity contrast can be simply

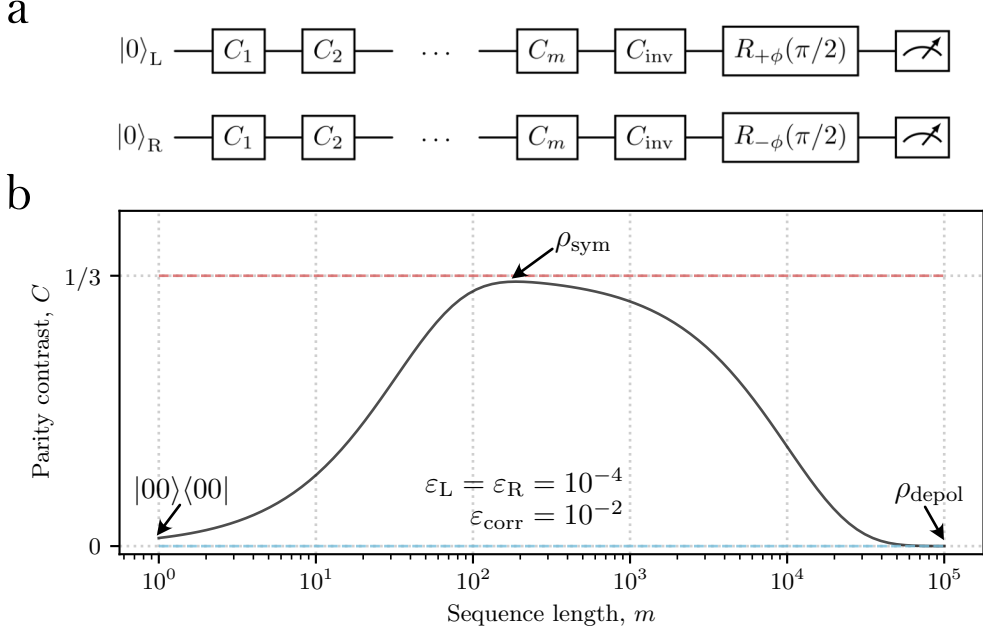


Figure 1.4: circuit diagram for parity measurement. Could there also be a nice schematic of the Bloch sphere picture of the differential rotation?

calculated by extending equation 1.44 to include the final differential rotation, in transfer matrix form, Λ_Π ,

$$\begin{aligned}\Pi(\phi) &= \langle \langle \pi | \Lambda_\Pi(\phi) \Lambda_T^m | 00 \rangle \rangle, \\ &= \mathcal{C} \cos(2\phi),\end{aligned}\tag{1.61}$$

where Π is the parity signal, $|\pi\rangle\rangle = |00\rangle\rangle + |11\rangle\rangle - |01\rangle\rangle - |10\rangle\rangle$, and $\mathcal{C}$ is the parity contrast. Evaluating $\Pi(\phi)$ for *common* and depolarising noise, we find the parity contrast follows,

$$\mathcal{C} = \frac{1}{3}(\alpha_{tr}^m - \alpha_{dt}^m),\tag{1.62}$$

with α_{tr} and α_{dt} given by equation 1.39. This method is demonstrated experimentally in section 1.7.3 below.

Furthermore, we may now isolate some α decay parameters by combining the contrast measurements with the computational state measurements made in *k-copy* RB. Namely, if we define the computational parity, Π_Z , as

$$\begin{aligned}\Pi_Z &= \mathcal{P}_{SS} + \mathcal{P}_{FF} - \mathcal{P}_{SF} - \mathcal{P}_{FS} \\ &= \frac{1}{3}(2\alpha_{dt}^m + \alpha_{tr}^m),\end{aligned}\tag{1.63}$$

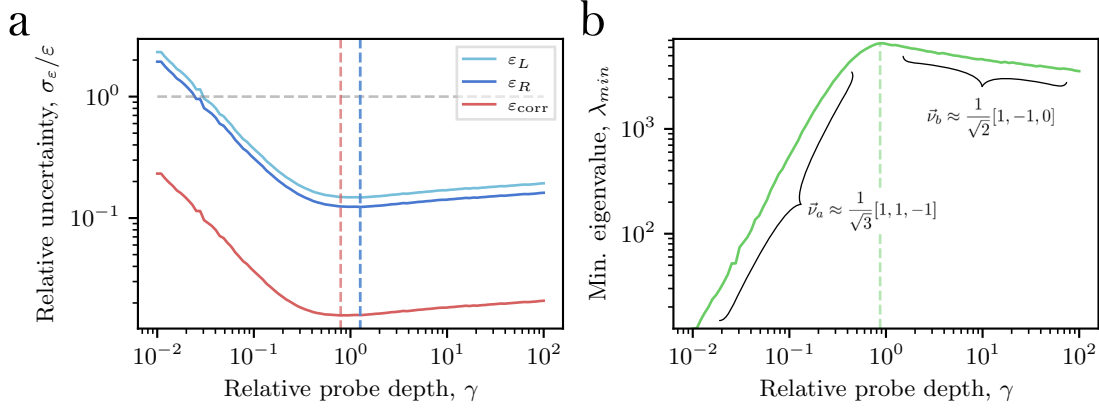


Figure 1.5: The performance of the *k-copy* fitting as a function of maximum Clifford sequence length (probe depth). Probe depth, γ , is relative to the characteristic decay length for the slowest mode, α_{tr}^m . Panel (a) uses the Cramér-Rao derived relative uncertainty for each extracted ε , while (b) shows the minimum eigenvalue of the Fisher information matrix, with associated eigenvectors highlighted.

then in combination with $\mathcal{C}$, we may isolate single exponential decays originating from the independent sectors named “symmetric trace”, and “diagonal traceless”. Specifically the combinations

$$\begin{aligned}\Pi_Z - C &= \alpha_{dt}^m, \\ \Pi_Z + 2C &= \alpha_{tr}^m.\end{aligned}\tag{1.64}$$

1.5 Sampling strategy

In this section we find the Fisher information gained from a single shot of the *k-copy* RB protocol, and use this to find a strategy for setting the sequence lengths, m . We then explore the shot efficiency of extracting error per Clifford parameters for a range of underlying noise magnitudes, using both the *k-copy* protocol described in section 1.3, and using the parity measurement protocol described in section 1.4.

The Fisher information, $\mathcal{I}$, quantifies what we can learn from random observable, X , about a set of underlying unknown parameters, θ , that models the probability of observing X . It is particularly informative to consider the Cramér-Rao bound,

which states that the inverse of $\mathcal{I}$ is a lower bound for the variance of any unbiased estimator of θ ,

$$\text{Cov}(\theta) \succeq \mathcal{I}(\theta)^{-1}. \quad (1.65)$$

The Fisher matrix is defined as,

$$\mathcal{I}_{ij}(\theta) = \mathbb{E}_X \left[\frac{\partial}{\partial \theta_i} \log f(X; \theta) \frac{\partial}{\partial \theta_j} \log f(X; \theta) \right], \quad (1.66)$$

where $f(X; \theta)$ is the likelihood function, and θ is a set of parameters. Here, we are interested in a multinomial probability function, with each shot resulting in a categorical outcome $X \in \{SS, SF, FS, FF\}$. The population functions given by equations 1.45, are then the probability of observing $X = x$ for a given set of parameters, $\theta = \{\varepsilon_L, \varepsilon_R, \varepsilon_{\text{corr}}\}$, yielding,

$$f(X = x; \theta) = P_x(\theta). \quad (1.67)$$

We now find the Fisher matrix to be,

$$\mathcal{I}_{ij}(\theta) = \mathbb{E}_X \left[\frac{1}{P_X^2} \frac{\partial P_X}{\partial \theta_i} \frac{\partial P_X}{\partial \theta_j} \right]. \quad (1.68)$$

For a discrete random variable, the expectation is,

$$\mathbb{E}[g(X)] = \sum_x P_x g(x), \quad (1.69)$$

giving our final expression for the single-shot multinomial Fisher matrix,

$$\mathcal{I}_{ij} = \sum_{x \in \{SS, SF, FS, FF\}} \frac{1}{P_x} \frac{\partial P_x}{\partial \theta_i} \frac{\partial P_x}{\partial \theta_j}. \quad (1.70)$$

The information gained in any one shot depends on the underlying error magnitudes, as well as the chosen sequence length, m . As the correlated profiles, P_x , appear as multi-exponentials decays dependent on common parameters, θ , we must choose a range of probe sequence lengths that yield large information over all the characteristic decays. As the characteristic decays are unknown, a good heuristic strategy for fitting an unknown exponential is logarithmic sampling **add citation here..** However, the total extent of sequence lengths to test is non-obvious. It would

be experimentally simple if the initial gradient of the correlated profiles successfully estimated the parameter θ , however, expanding the populations around $m = 0$, for small errors, we find that both purely uncorrelated and purely *common* noise decays are indistinguishable,

$$\mathcal{P}_{SS,LR}|_{m=0} \approx \mathcal{P}_{SS,\text{corr}}|_{m=0} \approx 1 - 2\varepsilon, \quad (1.71)$$

where we have assumed identical error magnitudes, $\varepsilon_L = \varepsilon_R = \varepsilon_{\text{corr}} = \varepsilon$. This suggests that we must collect data near the asymptotic structure of the decays to distinguish uncorrelated and *common* noise sources. To verify this, we set experimentally reasonable parameters for an underlying *common noise* dominated decay, and vary the maximum sequence length, m_{max} , while observing the expected uncertainty given by the Cramér-Rao bound of the Fisher information. We normalise by the magnitude of the parameter estimated to give the relative uncertainty, $\frac{\sigma_\varepsilon}{\varepsilon}$. We also find the minimum eigenvalue of the Fisher matrix, λ_{min} , which corresponds to how accurately the least constrained vector in parameter space can be measured. Figure 1.5 uses $N = 5000$ samples per sequence length (e.g. $s = 50$ shots, and $r = 100$ random sequences), and 10 total sequence lengths sampled logarithmically from $m_{\text{min}} > 2$ to $m_{\text{max}} = \gamma\tau$. The scalar γ is the relative probe depth and τ is the characteristic decay for the slow, α_{tr} mode,

$$\tau = \frac{-1}{\log(\alpha_{tr})} = \frac{-1}{\log(1 - 2(\varepsilon_L + \varepsilon_R))}. \quad (1.72)$$

We set error magnitudes of $\varepsilon_L = 1.0 \times 10^{-3}$, $\varepsilon_R = 1.2 \times 10^{-3}$, $\varepsilon_{\text{corr}} = 1.0 \times 10^{-2}$, which are similar magnitudes we observe experimentally (see section 1.7). It can be seen that the relative uncertainty for all θ are minimised when $\gamma \approx 1$. This length also corresponds to maximising λ_{min} — lowering the cost of estimating the least constrained vector of parameters, where the vector is written in the basis $[\hat{\varepsilon}_L, \hat{\varepsilon}_R, \hat{\varepsilon}_{\text{corr}}]$. It is interesting to note that the least constrained parameter vector, when $\gamma \ll 1$, roughly corresponds to $\vec{\nu} \approx \frac{1}{\sqrt{3}}[1, 1, -1]$, confirming our suspicion that *common* and depolarising noise is indistinguishable at small m . In the case that $\gamma \gg 1$, we find $\vec{\nu} \approx \frac{1}{\sqrt{2}}[1, -1, 0]$, corresponding to the depolarisation rates on the

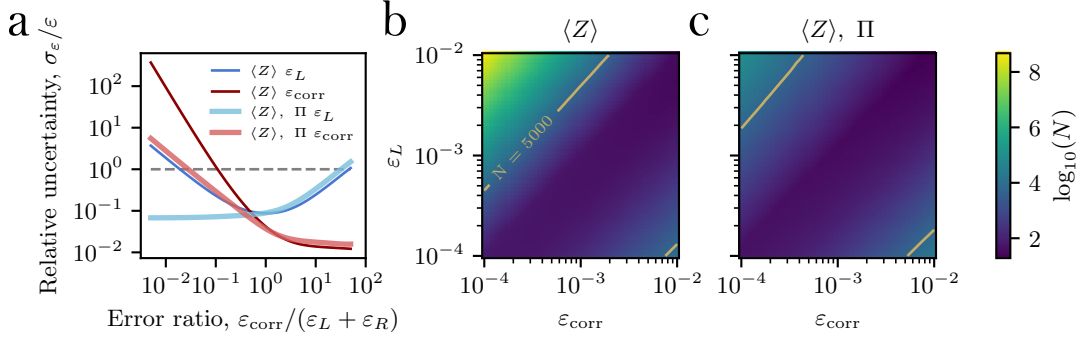


Figure 1.6: (a) The Cramér-Rao derived relative uncertainty for a range of correlated and uncorrelated error ratios. The thin lines correspond to expected uncertainties using $\langle Z \rangle$ measurements from *k-copy* RB, while thick lines use both $\langle Z \rangle$ and parity measurements, as described in section 1.4. (b-c) The required number of shots, N , per sequence length, m , for unity fractional uncertainty, for a range of correlated and uncorrelated errors. In (b), only $\langle Z \rangle$ measurements are used, while in (c) both $\langle Z \rangle$ and parity measurements are used. A contour is shown for 5000 samples, which is typically used in our experiments.

left and right qubit are tending to be indistinguishable, however we note that even over a couple of orders of magnitude, the distinguishability of ε_L and ε_R does not dramatically drop. Here, we draw the conclusion that a good strategy for efficient information per shot, is to set m_{max} to the characteristic decay of the slowest mode.

We now explore the ability for *k-copy* RB to fit θ for a range of underlying error magnitudes. Figure 1.6(a) shows the relative uncertainty as a function of the error ratio, defined here as $\varepsilon_{\text{corr}}/(\varepsilon_L + \varepsilon_R)$. We fix $\varepsilon_L = \varepsilon_R = 1 \times 10^{-3}$, and again set $N = 5000$ samples per sequence length m , with $m_{\text{max}} = \tau$, and vary $\varepsilon_{\text{corr}}$ to vary the error ratio. The thin blue and red line show the expected relative uncertainty for ε_L and $\varepsilon_{\text{corr}}$ respectively. We can see that for the case that local noise dominates, $\varepsilon_L \gg \varepsilon_{\text{corr}}$, the sufficient information for accurately estimating the correlated noise is not available. This limitation was found to be experimentally important, and motivates the injection of fixed *common noise* offsets in sections 1.7.4, and 1.7.6. Another strategy to alleviate this problem is by including data from the coherence probes described in section 1.4. The thick blue and red lines correspond to measurements where $N = 2500$ samples are collected by the typical $\langle Z \rangle$ *k-copy* protocol, and $N = 2500$ samples are using the parity protocol, denoted as Π . We

find that inclusion of the parity samples dramatically reduce the relative uncertainty of the fitted parameters when $\varepsilon_L \gg \varepsilon_{\text{corr}}$. Panels (b) and (c) show the required number of samples, N , for a relative uncertainty $\frac{\sigma_\varepsilon}{\varepsilon} = 1$, for a range of local and *common* noise magnitudes. We highlight the $N = 5000$ sample contour as this may be collected in reasonable time on our apparatus. Again, we see that including measurements using the parity protocol dramatically reduces the required number of samples in the regime of $\varepsilon_L \gg \varepsilon_{\text{corr}}$ (top left corner of plots).

Here we have shown that the Fisher information gathered in an experimentally realistic *k-copy* protocol is sufficient to accurately determine the underlying error per Cliffords, $\theta = \{\varepsilon_L, \varepsilon_R, \varepsilon_{\text{corr}}\}$, over many orders of magnitude of local and *common* errors. We shall show experimentally in section 1.7.4 that in the presence of experimental imperfections, and the fitting protocol (Levenberg-Marquadt), whose effects are not considered in this analysis, the proposed protocol still sufficiently recovers the magnitudes of the expected error per Cliffords with known noise injection.

1.6 Scaling to $k > 2$ qubits

The main text considers pairwise measurements to detect and quantify spatial correlations between qubits. This technique is inherently efficient and scalable to many qubits as single-shot survival populations of a k -qubit register can be collected simultaneously, with pairwise comparisons being made in a classical post-processing step. However, this limits the technique to detecting and quantifying only 2-body correlations. We define the weight of a crosstalk error as the number of qubits, w , comprising a subsystem that cannot be decomposed into a convex combination of single qubit channels, when the full system not under the action of the crosstalk channel, is a convex combination of single qubit channels. The weight of physical error channels highly depends on processor type and topology, however, it is suggested in Ref. [5], that many physically relevant spatial correlation errors are low-weight. As quantum computing platforms continue to scale, it will

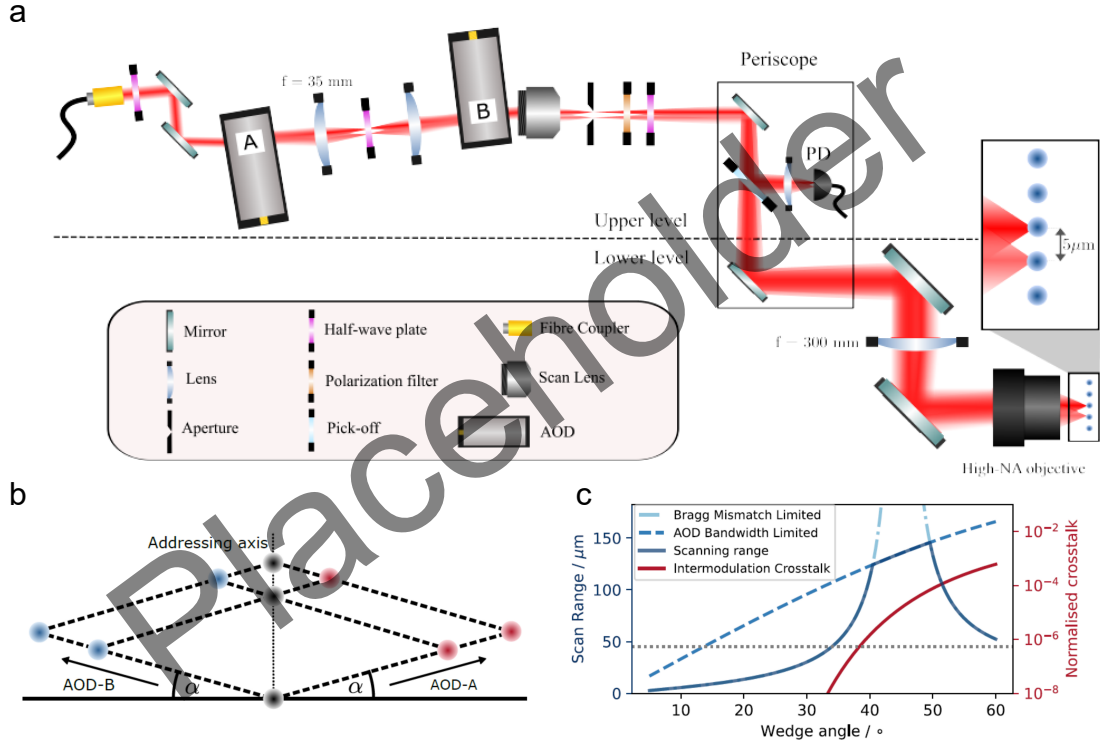


Figure 1.7: Asymptotic population decay for $k > 2$ qubits via numerical simulation.

soon become apparent to the engineer what error channels dominate in physical implementations.

We expect the symmetry resolved dynamics revealed by pairwise *k-copy* RB to generalise to higher weight errors, however, without prior work on the subrepresentation structure of the single-qubit Clifford group applied diagonally to > 2 qubits, extending the analysis for *k-copy* RB is challenging. However, for the *common noise* error model discussed throughout this work, a simple relation can be found for the asymptotic states. Namely, under *common noise* the state is driven to the maximally mixed state on the symmetric subspace, as this is the $\text{SU}(2)$ -invariant state on the symmetric irrep. This state is given by,

$$\rho_{\text{sym}} = \frac{1}{k+1} \sum_{l=0}^k |D_l^k\rangle \langle D_l^k|, \quad (1.73)$$

where $|D_l^k\rangle$ is the Dicke state for l excitations (here successful recoveries of target state), out of k -qubits. From this expression we see that an asymptote of

$\mathcal{P}_{SS\dots S}|_{m \rightarrow \infty} = 1/(k+1)$ is expected under *common noise*, while the maximally mixed state under local depolarisation leads to the familiar asymptote of $\mathcal{P}_{SS\dots S}|_{m \rightarrow \infty} = 1/(2^k)$. This result suggests that the presence of *common noise* affecting many qubits may be simply observed, however, without the explicit representation structure, it cannot be easily quantified.

For three-qubits, we leverage symbolic computation to diagonalise the resulting PTM representation of *common noise* and recover analytic decays that quantify spatial correlations without deriving the representation structure. As an example, we show in figure 1.13 experimental data for three-qubits subject to *common noise*, with an analytic fit parametrised by local depolarising channels for each qubit, and a single 3-body *common noise* channel (note that this simple model excludes 2-body correlations). Here, $\mathcal{P}_{[SSF]}$ and $\mathcal{P}_{[SFF]}$ represent the sum of the permutations of single- and two-qubit failures respectively. Explicitly, $\mathcal{P}_{[SSF]} = \mathcal{P}_{SSF} + \mathcal{P}_{SFS} + \mathcal{P}_{FSS}$.

$$\alpha_1 = (1 - \varepsilon_1)^m, \quad (1.74)$$

$$\alpha_2 = (1 - \varepsilon_2)^m, \quad (1.75)$$

$$\alpha_3 = (1 - \varepsilon_3)^m, \quad (1.76)$$

$$\beta = \alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3, \quad (1.77)$$

$$\alpha_{\text{corr1}} = (1 - \varepsilon_{\text{corr}})^m, \quad (1.78)$$

$$\alpha_{\text{corr2}} = (1 - 3\varepsilon_{\text{corr}})^m, \quad (1.79)$$

$$\alpha_{\text{corr3}} = (1 - 6\varepsilon_{\text{corr}})^m, \quad (1.80)$$

$$M_1 = \beta, \quad (1.81)$$

$$M_2 = 3(\alpha_1 + \alpha_2 + \alpha_3)\alpha_{\text{corr1}}, \quad (1.82)$$

$$M_3 = 2\beta\alpha_{\text{corr2}}, \quad (1.83)$$

$$M_4 = \frac{6}{5}\alpha_1\alpha_2\alpha_3\alpha_{\text{corr3}}, \quad (1.84)$$

$$M_5 = \frac{9}{5}\alpha_1\alpha_2\alpha_3\alpha_{\text{corr1}}, \quad (1.85)$$

$$\mathcal{P}_{SSS} = \frac{1}{8} + \frac{1}{24}(M_1 + M_2 + M_3 + M_4 + M_5), \quad (1.86)$$

$$\mathcal{P}_{FFF} = \frac{1}{8} + \frac{1}{24}(M_1 - M_2 + M_3 - M_4 - M_5), \quad (1.87)$$

$$\mathcal{P}_{[SSF]} = \frac{3}{8} + \frac{1}{24}(-M_1 + M_2 - M_3 - 3M_4 - 3M_5), \quad (1.88)$$

$$\mathcal{P}_{[SFF]} = \frac{3}{8} + \frac{1}{24}(-M_1 - M_2 - M_3 + 3M_4 + 3M_5), \quad (1.89)$$

Limitation here is not including weight-2 errors.

1.7 Experimental Results

The results presented here constitute the first project-data output from the newly commissioned *FastGates* apparatus. This work was both necessary for confirming the utility of the *k-copy* RB protocol, as well as pushing the automation of the apparatus for robust operation, useful for all future experimental projects.

1.7.1 Physical implementation

We demonstrate *k-copy* RB using up to eight trapped $^{40}\text{Ca}^+$ ions separated by an average spacing of $\sim 5 \mu\text{m}$. The qubit is encoded in the optical transition $|4S_{1/2}, m_j = -\frac{1}{2}\rangle \leftrightarrow |3D_{5/2}, m_j = -\frac{5}{2}\rangle$ in a quantising field of $\sim 5 \text{ G}$. Single-qubit rotations are driven by 729-nm light, with details given in section ???. The two-qubit coherence and Stark shift measurements use the individual addressing system, detailed in section ??, while all other measurements use the global elliptical beam, detailed in section ??.

The single-qubit Clifford gate set is decomposed into native $\pi/2$ single-qubit rotations of the form $\hat{R}_x(\pi/2)$ and $\hat{R}_z(\pi/2)$: $\hat{\sigma}_x$ rotations are performed natively on resonance with the qubit transition, with relative phase set by modulating the 729-nm beam via an acousto-optic modulator (AOM), while $\hat{\sigma}_z$ rotations are performed virtually by updating the phase of subsequent $\hat{\sigma}_x$ rotations. Two decomposition strategies are used, crucially with varying average numbers of physical $\hat{\sigma}_x \pi/2$ -gates per Clifford. The first uses an average of $\kappa = 2.17 \pi/2$ -gates, while the second uses $\kappa = 1.167$

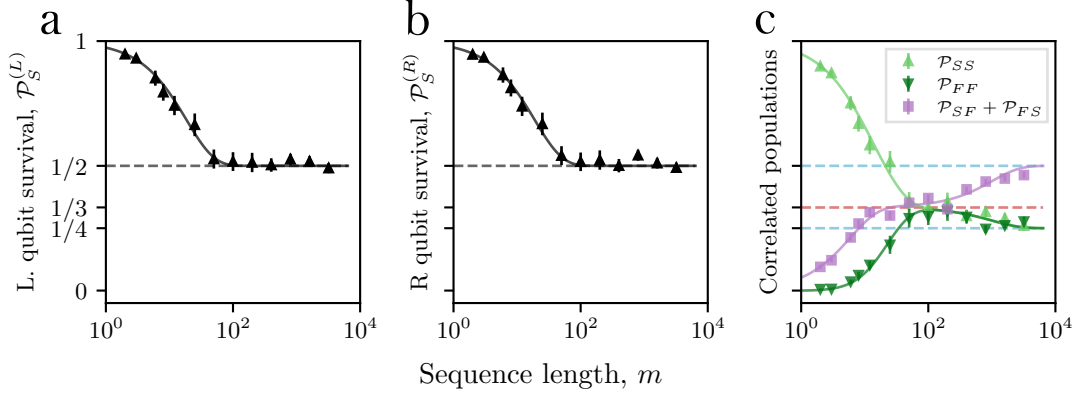


Figure 1.8: Correlated two-qubit probabilities after *k-copy* RB. Panels (a,b) show the single-qubit success probabilities, which decay with a single exponential (solid line) toward $\mathcal{P}_S|_{m \rightarrow \infty} = 1/2$. Panel (c) shows the correlated two-qubit populations, which exhibit an intermediate $\mathcal{P}_{SS} = 1/3$ asymptote before relaxing toward $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/4$.

gates.

We begin by reproducing the multi-exponential decays seen in simulation, figure 1.2, followed by parity measurements to observe the coherence of the intermediate symmetric state described in section 1.4. We then present data with known injected errors through phase and detuning offsets to show the ability for *k-copy* RB to discern correlated and uncorrelated errors for a range of magnitudes. We finish this section with evidence for scaling the protocol to more than two qubits. We show both data for three-qubit decay profiles, and pairwise analysis of an eight-ion chain with dominant correlated errors due to global drive field inhomogeneity.

1.7.2 Z-basis measurements

The *k-copy* RB protocol, with circuit diagram shown in figure 1.1, is executed on two qubits using a global field. To ensure a visually clear intermediate asymptote is seen, we set a fixed 10 kHz frequency detuning of the 729-nm laser compared to the bare qubit. As single qubits gates are calibrated for on-resonance pulses, this detuning will be a large *common noise* source. The Clifford sequence length is log-sampled up to $m = 3200$ total gates to capture both the intermediate asymptote due to this common unitary error, and the final decay to the fully depolarised state. The recovered multi-exponential decay from the correlated success and failure rates

of the two qubits is shown in figure 1.8(c). Each solid point is inferred through $r = 50$ randomisations and $s = 50$ shots per randomisation. **Move section above?** For each random sequence j , the population $\mathcal{P}_{SS,j}$ is estimated from the s binary measurement outcomes, $\mathcal{P}_{SS,j} = \frac{1}{s} \sum_{i=1}^s X_{ij}$, where $X_{ij} = 1$ if the i th shot is measured in state SS , and $X_{ij} = 0$ otherwise. The recovered population is then the mean over the r random sequences, $\mathcal{P}_{SS} = \frac{1}{r} \sum_{j=1}^r \mathcal{P}_{SS,j}$. Error bars show the standard error of the mean across random sequences,

$$\text{SE}(\mathcal{P}_{SS}) = \sqrt{\frac{1}{r^2} \sum_{j=1}^r (\mathcal{P}_{SS,j} - \mathcal{P}_{SS})^2}. \quad (1.90)$$

The correlated populations exhibit a clear intermediate asymptote at $m \approx 100$ Cliffords, corresponding to the initial state $|00\rangle$ approaching ρ_{sym} , equation ??, with $\mathcal{P}_{SS} \rightarrow \frac{1}{3}$ asymptote highlighted by the red dashed line. The dashed blue line shows the fully depolarised $\mathcal{P}_{SS} \rightarrow \frac{1}{4}$ asymptote, seen clearly by $m \approx 1000$ gates. The solid lines are found through modelling errors by a combination of local depolarising errors and *common noise*, given by equation 1.45 generalised to include independent state-preparation and measurement error (SPAM), ε_{sp} (see equation. 1.58 for example). The model is fit using the non-linear Levenberg–Marquardt method, yielding extracted parameters $\varepsilon_{sp} = 0(5) \times 10^{-3}$, $\varepsilon_{\text{corr}} = 2.6(2) \times 10^{-2}$, $\varepsilon_L = 5(6) \times 10^{-4}$, and $\varepsilon_R = 0(6) \times 10^{-4}$, confirming that correlated errors dominate with this global field detuning.²

From the same shot data record we recover the single-qubit populations required for standard RB, as described in section ??. To match the naming of the relevant participating irreps, we label the two qubits L and R for left and right respectively. The standard RB single exponential decay curves are shown in (a) and (b), with extracted single-qubit error per Cliffords of $\varepsilon_c^{(L)} = 2.6(2) \times 10^{-2}$ and $\varepsilon_c^{(R)} = 2.6(2) \times 10^{-2}$. As expected, *k-copy* RB provides evidence for this standard error per Clifford to be decomposed to local and correlated components, given by equation 1.51.

²The large relative uncertainty of both local depolarising rates is due to the large covariance between ε_L and ε_R . The fitting may be constrained to the sum of local errors, to find $\varepsilon_L + \varepsilon_R = 5(1) \times 10^{-4}$. However, if precision measurements of this local error is of interest, either r , and s could be increased, or the local error contributions could be isolated as discussed in section ??.

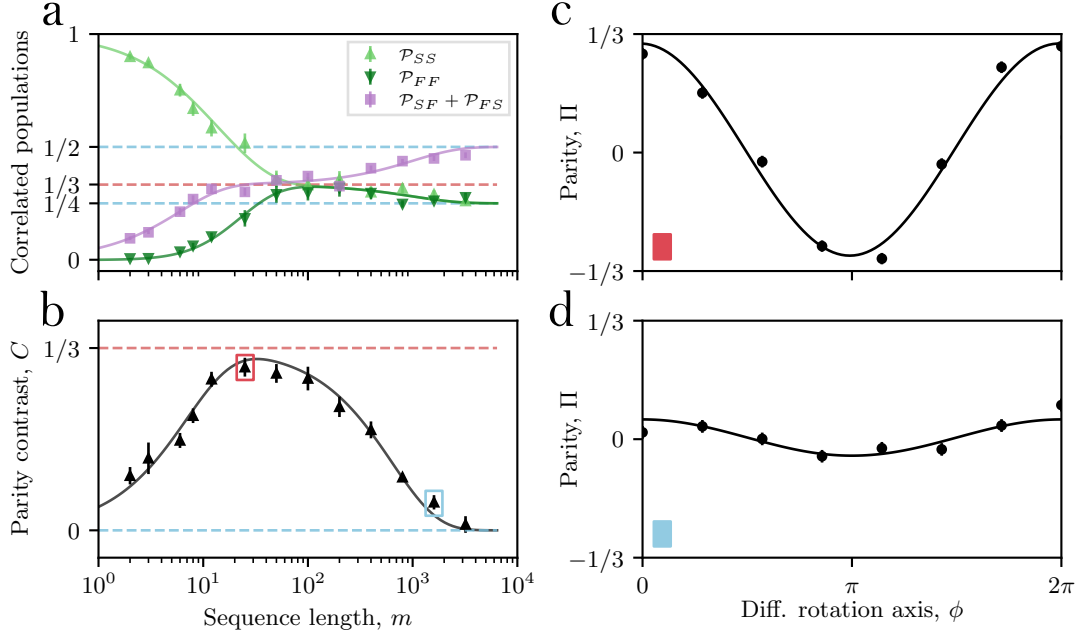


Figure 1.9: Parity contrast versus sequence length; (a) shows the standard Z measured correlated populations. (b) shows parity contrast — a signal proportional to occupation of the symmetric-trace sector. Solid lines are multi-exponential decays given by equation 1.45 and equation 1.62 for (a) and (b) respectively. Dashed lines indicate the expected asymptotes. (c) and (d) show example parity fringes for sequences of $m = 25$ and $m = 1600$ Clifford gates respectively with solid lines given by equation 1.61.

1.7.3 Coherence measurements

A partial-tomography parity measurement, given by the circuit seen in figure 1.4(a), is used to independently confirm occupation of the symmetric sector, quantifying overlap of the final state with Bell state, $|\Psi^+\rangle$. As the final differential rotation pulse requires local addressing of the ions, the single addressing system is used. Each random sequence is repeated for varying final rotation angle, ϕ , to produce parity fringes, examples shown in (c) and (d) in figure 1.9 for $m = 25$ and $m = 1600$ Clifford gates respectively. The contrast is extracted by equation 1.62, and a plot against sequence length is shown in (b). The contrast asymptote $\mathcal{C} \rightarrow 1/3$ is observed for $m \in [10, 100]$, corresponding to the residual coherence of ρ_{sym} , equation 1.53. The parity contrast data is fit (solid line) according to equation 1.62 to extract error per Cliffords of, $\varepsilon_{\text{corr}} = 2.1(2) \times 10^{-2}$ and $\varepsilon_L = \varepsilon_R = 3.9(3) \times 10^{-4}$.

1.7.4 Controlled noise injection

To verify the ability for *k-copy* RB to discern correlated and uncorrelated errors, we compare the magnitude of extracted errors to known injected errors. We limit the injected error method to those consisting of noise on common drive fields, and local depolarising type noise. Both of these noise models can be induced entirely through single qubit rotations, which are trivial to implement on our device.

Laser phase jitter

We first inject correlated errors via phase noise on a common drive field. This noise mechanism is prevalent in many platforms that use a common oscillator, such as laser, microwave, or radio-frequency (RF) fields to resonantly drive qubit transitions.

The injected error magnitude is inferred through considering the ideal and noisy unitary channel. A unitary qubit rotation is given by,

$$\hat{U} = e^{-i\frac{\theta}{2}\vec{n}\cdot\vec{\sigma}}, \quad (1.91)$$

where θ is the rotation angle, and $\vec{n} \cdot \vec{\sigma}$ is the rotation axis. We decompose the single-qubit Clifford gate set into a series of resonant $\theta = \pi/2$ qubit rotations around axes $\hat{\sigma}_x$ and $\hat{\sigma}_z$. Practically, phase noise on either the qubit frequency or drive field will introduce some deviation of this rotation axis. We shall consider the effect of zero-mean phase noise on $\hat{\sigma}_x$ rotations, leading to an erroneous rotations axis,

$$\hat{\sigma}_{\delta\phi} = \cos(\delta\phi)\hat{\sigma}_x + \sin(\delta\phi)\hat{\sigma}_y, \quad (1.92)$$

where $\delta\phi$ is the standard deviation of the noise in radians. We may find the expected error of a single Clifford gate using,

$$\varepsilon_{\delta\phi} = \kappa(1 - \mathcal{F}(V, U)), \quad (1.93)$$

where κ is the average number of $\pi/2$ rotations in a Clifford gate, and $\mathcal{F}(V, U)$ is the average gate fidelity. The fidelity of two unitaries averaged over the Haar measure is given in appendix equation A.3, reproduced here,

$$\mathcal{F}(V, U) = \frac{1}{d(d+1)} \left(|\text{Tr}(U^\dagger V)|^2 + d \right). \quad (\text{A.3})$$

Considering $\theta = \pi/2$ rotations, we find,

$$\hat{U} = e^{-i\frac{\pi}{4}\hat{\sigma}_x} = \frac{1}{\sqrt{2}} \left(\hat{I} - i\hat{\sigma}_x \right), \quad (1.94)$$

$$\hat{V} = e^{-i\frac{\pi}{4}\hat{\sigma}_{\delta\phi}} = \frac{1}{\sqrt{2}} \left(\hat{I} - i\hat{\sigma}_{\delta\phi} \right). \quad (1.95)$$

The fidelity, given that dimension $d = 2$ for a single qubit, is found to be,

$$\mathcal{F}(V, U) = \frac{1}{6} \left((1 + \cos(\delta\phi))^2 + 2 \right). \quad (1.96)$$

For small phase noise standard deviations, $\delta\phi \ll 1$, we find an expected error per Clifford of,

$$\varepsilon_{\delta\phi} = \frac{\kappa}{3} (\delta\phi)^2. \quad (1.97)$$

A known phase error may be conveniently injected through phase offsets sampled from a normal distribution with zero mean and standard deviation $= \delta\phi$ applied to the RF drive of a control AOM on the 729-nm global addressing path. The phase of the AOM RF is imprinted directly onto the 729-nm light leading to the expected drop in fidelity of the Clifford gates. In this experiment $\kappa = 2.17 \pi/2$ rotations. A full *k-copy* RB experiment is executed with logarithmic sampling up to $m = 512$ Clifford gates with $r = 50$ randomisations and $s = 50$ repeats. The resulting multi-exponential success and failure probability decay curves are fit to equation ??, again generalised to include state-preparation errors. The correlated and local error components are found for a range of injected phase noise standard deviations. The resulting measured errors are shown in figure 1.10. The x -axis is the expected error injected $\varepsilon_{\delta\phi}$, while the y -axis is the extracted error magnitudes. The fit, shown by the solid red line, is a direct proportionality of injected error plus a baseline correlated error rate not due to phase injection,

$$\varepsilon_{\text{corr}} = \varepsilon_{\text{corr},0} + \varepsilon_{\delta\phi}. \quad (1.98)$$

The baseline correlated error is therefore the only parameter to fit. With $\text{varepsilonpsilon}_{\text{corr},0} = 1.71(4) \times 10^{-3}$, we find good agreement between the model and data. The local depolarising error on both qubits remains relatively constant, with the left qubit

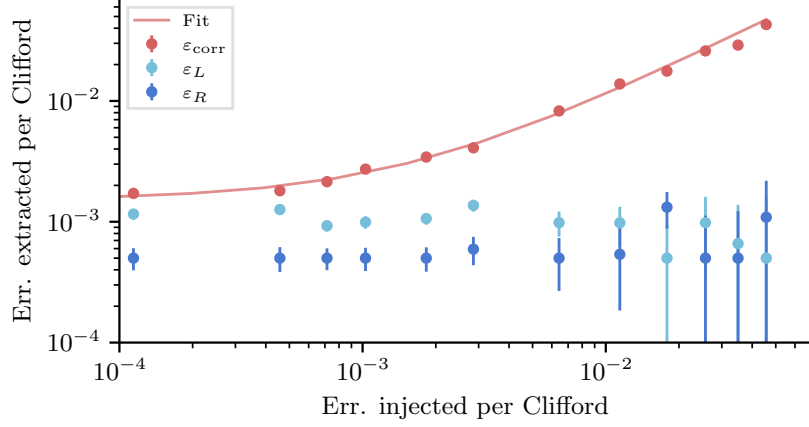


Figure 1.10: Injecting correlated noise between two qubits via common drive phase noise. Points correspond to fitted error per Clifford from full *k-copy* RB experiments. The solid red line is a linear fit between measured and injected error.

having a larger local error compared to the right qubit. We believe this asymmetry in local error is due to the global 729-nm beam not being centrally aligned to both qubits. The uncertainty in extracted error is found through Levenberg-Marquadt non-linear fit, it can be seen that the uncertainty in the errors increases with the total correlated error. **Compare this with analytic result when multi-exps are of different magnitudes.**

Laser detuning scan

k-copy RB can isolate errors arising from shared control fields, making it useful both for device characterisation and for optimising global control parameters. A fixed detuning of the global drive, under the action of *k-copy* Clifford twirling, is well modelled by the common random unitary error channel described in section 1.3.3. We demonstrate this by scanning the 729-nm laser frequency used for single-qubit rotations away from the bare qubit resonance. As with the previous section, we consider the fidelity of the error channel compared to the ideal $\pi/2 \hat{\sigma}_x$ single qubit rotation.

A coherent detuning error is written as,

$$V = \exp \left(-i \frac{\pi}{4} \left(\hat{\sigma}_x + \frac{\delta}{\Omega} \hat{\sigma}_z \right) \right), \quad (1.99)$$

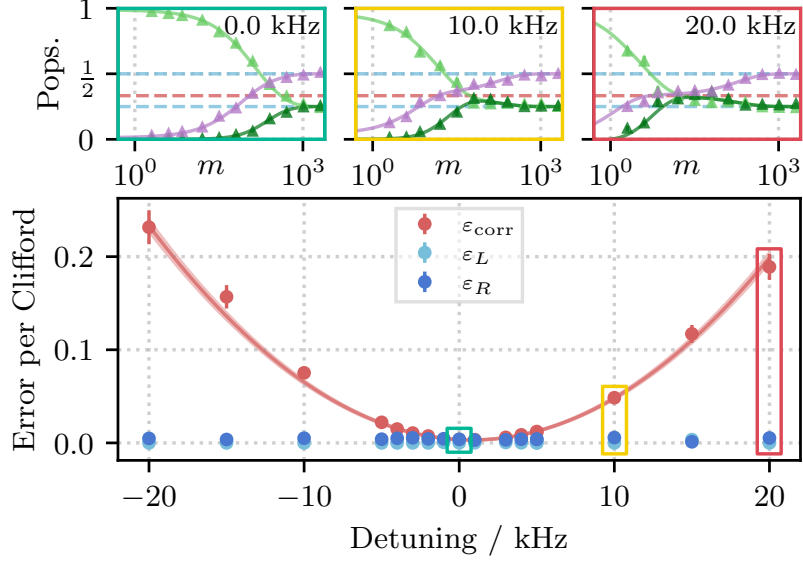


Figure 1.11: Remake figure for thesis. Correlated and single-qubit depolarising errors extracted as a function of shared-drive detuning. The correlated error (red points) rises quadratically with laser detuning, while the local depolarising errors (blue points) remain approximately constant. Error bars are the extracted standard deviations from a non-linear Levenberg–Marquardt fit of the underlying correlated populations. The red shaded region signifies the 68% confidence, generated through covariance-based parameter sampling.

where δ is the detuning from resonance, and Ω is the Rabi frequency. Assuming $\delta/\Omega \ll 1$, the fidelity is found to be,

$$\mathcal{F}(V, U) = 1 - \frac{1}{3} \left(\frac{\delta}{\Omega} \right)^2 + \mathcal{O} \left(\frac{\delta}{\Omega} \right)^4. \quad (1.100)$$

To convert this to an error contribution to a single qubit Clifford gate, we introduce the scalar κ_δ , to account for both the number of $\pi/2$ rotation in each Clifford gate, but also the duration and delays between pulses where phase tracking between qubit and laser also drift. The error contribution is then given by

$$\varepsilon_\delta = \kappa_\delta (1 - \mathcal{F}(V, U)) = \frac{\kappa_\delta}{3} \left(\frac{\delta}{\Omega} \right)^2 + \mathcal{O} \left(\frac{\delta}{\Omega} \right)^4. \quad (1.101)$$

Each data point is obtained from a complete *k*-copy RB experiment with 11 sequence lengths up to $m = 1024$, using $r = 50$ random sequences and $s = 50$ shots per sequence, with correlated populations fit to equation ?? to extract $\{\varepsilon_{\text{corr}}, \varepsilon_L, \varepsilon_R\}$ at each detuning. The upper panels of figure 1.11 show example datasets for the

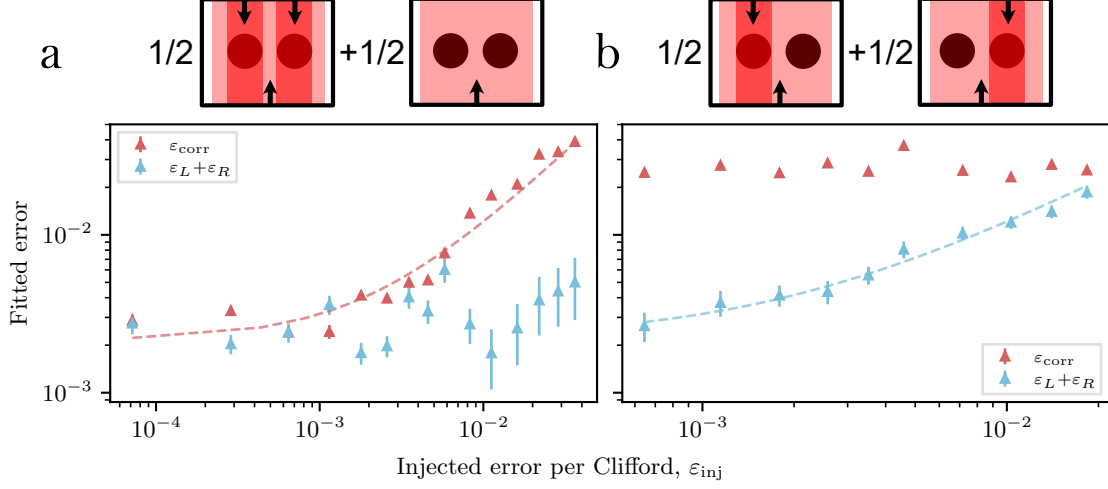


Figure 1.12: Correlated and uncorrelated errors are injected via a single-addressed Stark-shift beam, and quantified through fitting the correlated success populations. The four addressing configurations are depicted visually above the plot. The wide beam signifies the on-resonance global drive which executes the parallel single-qubit rotations, while the narrow beams represent the off-resonant Stark shift beams addressing single-qubits (black circles). In (a) the errors are correlated between qubits, while panel (b) shows the extracted errors after independent errors are injected. Error bars are the extracted standard deviations from a non-linear Levenberg–Marquardt fit of the underlying correlated populations.

k-copy RB sequences, while the lower panel shows the extracted error per Cliffords as a function of detuning. The correlated error is observed to increase quadratically with detuning, while the local depolarising errors remain approximately constant, consistent with detuning of a shared drive producing *common noise*. To fully parameterise the injected error, we include a background correlated error not originating from detuning, ε_0 , and a shift of the light-shifted qubit versus the bare qubit frequency, δ_0 . The solid red line is therefore fit to the model,

$$\varepsilon_{corr} = \varepsilon_0 + \frac{\kappa_\delta}{3} \left(\frac{\delta_L - \delta_0}{\Omega} \right)^2. \quad (1.102)$$

Here, $\Omega = 2\pi \times 100(1)$ kHz is the independently measured Rabi frequency. Fitting gives $\varepsilon_0 = 3.2(1) \times 10^{-3}$, $\kappa_\delta = 15.9(7)$, and $\delta_0 = 2\pi \times 0.80(5)$ kHz.

Single addressed stark shifts

Uncorrelated noise injection requires the introduction of noise with spatial inhomogeneity. This is implemented experimentally through an off-resonant single addressing beam inducing Stark-shifts between the on-resonant parallel single qubit rotations executing the *k-copy* twirling. The single addressing beam is detuned from the qubit resonance by -15 MHz, and the induced Stark shift with beam power, γ , is calibrated using a Ramsey sequence, here measured as $\gamma = 8.69 \times 10^7 \text{ Hz W}^{-1}$ **uncertainty?**. The injected error, ε_{inj} , is inferred by,

$$\varepsilon_{\text{inj}} = \frac{\kappa}{3} \phi^2, \quad (1.103)$$

where the phase error, given in radians is, $\phi = P\tau\gamma$, with P being the power in the single addressing beam, τ the duration of the Stark-shift pulse. In these experiments, the duration of the Stark-shift pulse is fixed to $\tau = 20 \text{ } \mu\text{s}$, while the power of the beam is varied to supply a known error.

Two experimental sequences were run, the first, shown in figure 1.12(a), implements stochastic *common noise* through applying either two spots shifting both ions, or no Stark shift, with a 50 : 50 chance after each Clifford gate. The second experiment, shown in (b), applies local errors through a single spot either addressing the left or right qubit, again with 50 : 50 chance after each Clifford gate. These two arrangements are shown schematically in the upper panels. To ensure correlated errors dominate even with injected local errors, an additional constant correlated error offset of $\varepsilon_{\text{corr},0} \approx 2 \times 10^{-2}$ is applied through random phase fluctuations, as described in section 1.7.4. This reduces the number of samples required to accurately estimate the correlated and uncorrelated error contributions, which is of practical importance in collecting data in reasonable time (for reference, this data was collected over ~ 3 hours). It can be seen that $\varepsilon_{\text{corr}}$ increases only when injecting *common noise*, and remains constant when local errors are injected. Dashed lines are simple linear fits given by $\varepsilon_{\text{fit}} = \varepsilon_0 + \varepsilon_{\text{inj}}$, where ε_{fit} are the extracted errors, ε_0 is some residual error baseline. These straight line fits show good agreement between measured and injected error when accounting for some fit baseline error rates.

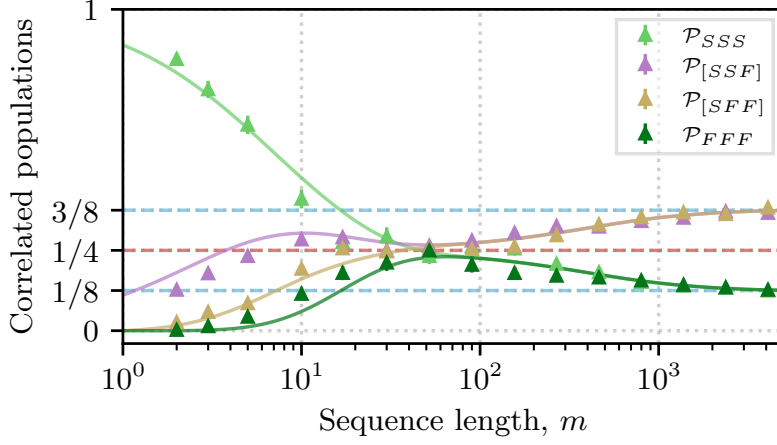


Figure 1.13: Experimental data and analytic fit for 3-qubits subject to correlated random-unitary single-qubit noise injected via random phase jitter ($\sigma_\phi = 0.06$ turns).

1.7.5 Three-qubit decay

As an example, we show in figure 1.13 experimental data for three-qubits subject to *common noise*, with an analytic fit parametrised by local depolarising channels for each qubit, and a single 3-body *common noise* channel (note that this simple model excludes 2-body correlations). Here, $\mathcal{P}_{[SSF]}$ and $\mathcal{P}_{[SFF]}$ represent the sum of the permutations of single- and two-qubit failures respectively. Explicitly, $\mathcal{P}_{[SSF]} = \mathcal{P}_{SSF} + \mathcal{P}_{SFS} + \mathcal{P}_{FSS}$.

1.7.6 Scaling to a multi-ion register

Pairwise *k*-copy RB extends simply to > 2 qubits: correlated populations are fit to equation 1.45 independently for each qubit pair. For a register of k qubits, all $k(k-1)/2$ pairwise outcomes are extracted from the same shot record — no additional trials on the quantum register are required, although classical post-processing scales with the number of pair combinations. The protocol itself requires only synchronised application of identical single-qubit Clifford sequences and spatially resolved readout.

This scaling is demonstrated on an eight-ion chain, with parallel single-qubit rotations driven by an elliptical 729-nm beam. We expect to observe structured

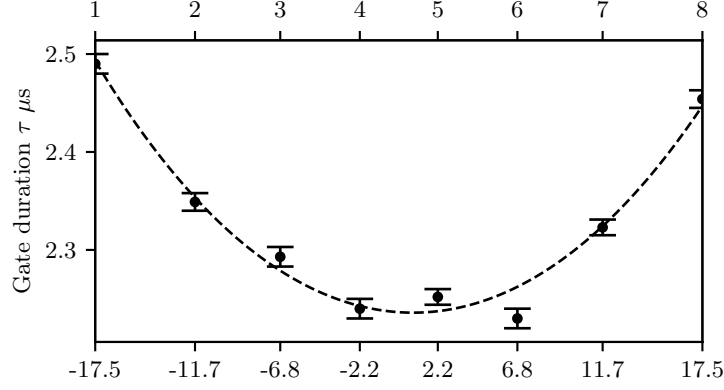


Figure 1.14: Variation of π -pulse gate duration along eight ion chain due to global beam Gaussian profile.

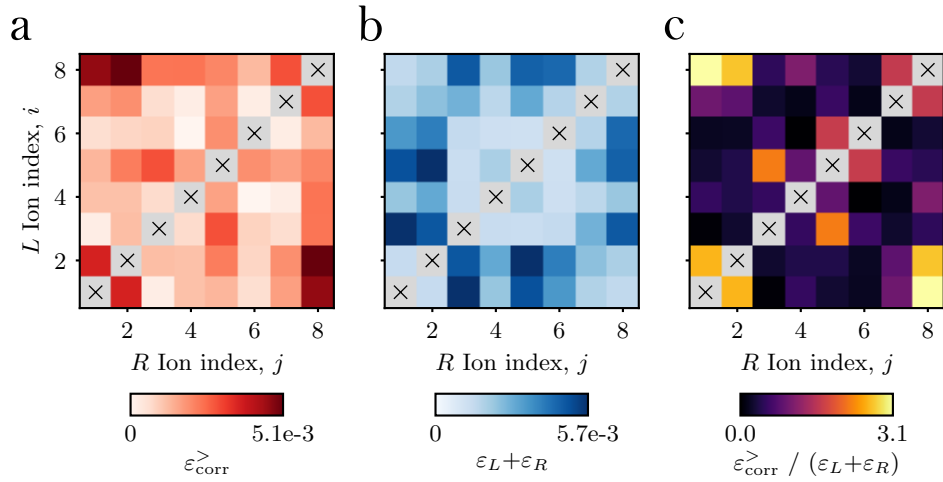


Figure 1.15: Correlated and local error per Clifford measured for all qubit pairs after applying *k-copy* RB to an 8 ion chain. Noise is dominated by position dependent duration of $R_x(\pi/2)$ gates consistent with inhomogeneous illumination by the global drive. Panel (a) displays the extracted correlated errors, $\varepsilon_{\text{corr}}^{(i,j)}$ for each qubit pair, (i, j) . Panel (b) displays the sum of both independent depolarising errors, $(\varepsilon_L + \varepsilon_R)^{(i,j)}$. Panel (c) displays the ratio of excess correlated noise (offset from phase noise injection), $(\varepsilon_{\text{corr}}^>)^{(i,j)} = \varepsilon_{\text{corr}}^{(i,j)} - \varepsilon_{\text{corr}}^{(0)}$ to the sum of local errors given in (b). This plot highlights the outer qubits are dominated by spatially correlated errors.

errors due to the Gaussian intensity profile of the beam. To verify the expected gate duration dependence due to the finite beam, we run parallel gate duration calibration experiments (see section ?? for details). The gate duration, τ , as a function of qubit position in the beam is expected to follow,

$$\tau(r) \propto \frac{1}{\sqrt{I(r)}}, \quad (1.104)$$

where $I(r)$ is a Gaussian distribution, $I = a \exp\left(\frac{-(r-r_0)^2}{2c^2}\right)$. Figure 1.14 shows the measured π -pulse gate durations for each ion, with the fit given by the dashed line. From this, we find an expected major-axis waist radius of 27.7(9) μm aligned to the ion chain. The qubit spacing, on the lower axis of the calibration plot, is inferred through the axial confinement of 0.5 MHz using equilibrium positions of a linear chain given in [24]. At this confinement, the chain extent is 35 μm , giving an expected Rabi-frequency difference of 10.3(6)% between the outer and inner ions. We calibrate global single-qubit gates using the inner ion, $i = 4$, leading to an expected per Clifford error of (using fidelity definition from appendix A.5),

$$\varepsilon_{c,max} = \frac{2}{3} \kappa \left(\alpha \frac{\pi}{4} \right)^2 = 5.08(2) \times 10^{-3}, \quad (1.105)$$

where, here the Clifford decomposition used gives $\kappa = 1.167$, and the fractional Rabi frequency difference is given by the above calibration curve, $\alpha = 0.103(6)$.

As with the single qubit Stark-shift dataset, we ensure that correlated errors dominate to improve the fitting sensitivity. This is done through a fixed phase noise error corresponding to $\varepsilon_{corr,0} = 1.3 \times 10^{-2}$. In the following data, we subtract this baseline offset to highlight the excess correlated noise due to the beam structure,

$$\varepsilon_{corr}^> = \varepsilon_{corr} - \varepsilon_{corr,0}.$$

Figure 1.15 shows the extracted pairwise correlations. The excess correlated error is shown in (a), with the largest values seen for the outer ion pairs, agreeing with the expected maximum error of $\varepsilon_{corr,max}^> = \varepsilon_{c,max}$. Panel (b) shows the sum of the local depolarisation error per Cliffords on both qubits, with the largest values now corresponding to pairs consisting of one central ion, and one outer ion. We plot the ratio of excess correlated error and depolarisation errors in panel (c) to further highlight the error distribution due to the beam shape.

1.8 Conclusions and outlook

k-copy RB efficiently discerns correlated and uncorrelated error channels acting on a multi-qubit system. Because of the symmetry enforced by twirling over the diagonal Clifford representation $Cl^{\otimes k}$ (Section ??), structured errors drive the decay of correlated populations into invariant subspaces in a way that is directly observable without full process reconstruction. Given a specified error model, these correlated decays can be used to directly quantify the magnitude of the individual error channels contributing to the total error per Clifford.

This method has been proposed and experimentally demonstrated, and its application extended to an eight-qubit register through pairwise comparisons (Section 1.7.6). This approach permits highly efficient parallel data collection: the time required on the device is independent of qubit number, and the protocol requires exclusively local operations followed by classical post-processing of correlated survival rates.

Expand this Outlook section substantially for the thesis, drawing together the forward-looking threads seeded earlier in this chapter and in Chapter ??:

- **Extension to genuine ≥ 3 -body correlated error terms, beyond the pairwise diagnostic of Section 1.7.6.**
- **Experimental characterisation of ZZ -type entangling errors (Section ??), not demonstrated here.**
- **Sample-cost / Fisher-information analysis for multi-exponential fitting — flagged as an open numerics question in the manuscript SM and not yet resolved.**
- **Non-Markovian (temporally correlated) noise, via repeated identical Clifford RB sequences applied at different times on a single qubit, exploiting the same correlated-outcome analysis used here for spatial correlations.**

- Character-RB-style initial-state engineering (Section ??) to isolate single exponential decays and reduce sample cost.

Appendices

A

Appendix

A.1 Generating Ions

A.2 Extracting Laser Offset and Magnetic Field

A.3 Experimental Control

A.4 Creating Squeezed States

A.5 Gate Fidelity

We derive the fidelity of a noisy gate implementation with respect to the ideal unitary.

A quantum channel,

$$\mathcal{E} : \rho \mapsto \rho, \tag{A.1}$$

which executes ideal unitary, U , with some error, has an average operation fidelity given by,

$$F(\mathcal{E}, U) = \int d\psi \langle \psi | U^\dagger \mathcal{E}(|\psi\rangle \langle \psi|) U | \psi \rangle, \tag{A.2}$$

where the integral is taken over the uniform Haar measure on the space of pure states. We shall see that the language used for this analysis of single qubit

gate fidelities closely resembles the group theory approach describing randomised benchmarking. The Haar measure is relevant here to consider the fidelity of the operation independent of the initial state (assuming it is pure), which is exactly the enforced criteria through Clifford twirling during RB. For a single qubit, this can be expressed as an integral over the Bloch sphere. If we further assume that the error channel is unitary, as is the case of a detuned channel considered here, we shall find that the fidelity follows

$$F(V, U) = \frac{1}{d(d+1)} \left(|\text{Tr}(U^\dagger V)|^2 + d \right), \quad (\text{A.3})$$

where V is the error unitary, and d is the dimension of the Hilbert space.

Here we shall give a derivation of this result, by considering the symmetry imposed by the Haar measure. The standard fidelity of two unitaries following from equation A.2 is given by

$$\begin{aligned} F(V, U) &= \int d\psi \langle \psi | U^\dagger V | \psi \rangle \langle \psi | V^\dagger U | \psi \rangle, \\ &= \int d\psi |\langle \psi | W | \psi \rangle|^2, \end{aligned} \quad (\text{A.4})$$

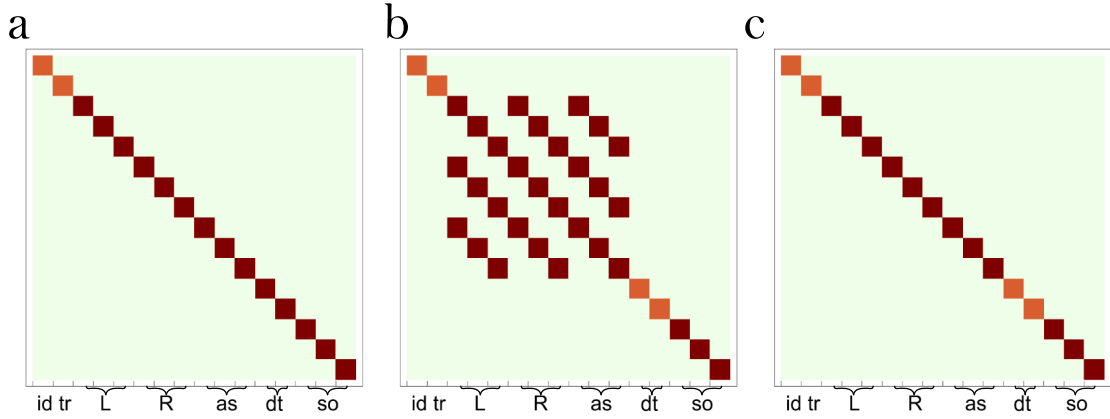
where $W = U^\dagger V$. Using the identity $\text{Tr}(A)\text{Tr}(B) = \text{Tr}(A \otimes B)$, we can rewrite the fidelity as a two-copy expectation value,

$$F(V, U) = \text{Tr} \left((W \otimes W^\dagger) \int d\psi (|\psi\rangle \langle \psi|)^{\otimes 2} \right), \quad (\text{A.5})$$

where we have now separated the fidelity into a part dependent on the error channel, and a part equivalent to the second moment of the Haar measure, explicitly rewritten as

$$\mathcal{M}_2 = \int d\psi (|\psi\rangle \langle \psi|)^{\otimes 2}. \quad (\text{A.6})$$

We now consider two apparent symmetries of $\mathcal{M}_2$: first, the Haar measure is invariant under unitary transformations, and so our expression must commute with $U \otimes U$ for any unitary U ; second, the two states in the tensor product are identical, and so $\mathcal{M}_2$ must be invariant under the swap operator S which exchanges the two copies. These

Figure A.1: **ZZ-noise**

two symmetries are sufficient to show that $\mathcal{M}_2$ is proportional to the projector onto the symmetric subspace of the two-copy Hilbert space, which is given by

$$\Pi_{\text{sym}} = \frac{1}{2}(\mathbb{I} + S). \quad (\text{A.7})$$

To normalise the moment (as the Haar measure is a probability distribution), we enforce $\text{Tr}(\mathcal{M}_2) = 1$. In this two-copy expression, it is apparant that $\text{Tr}(I) = d^2$ and $\text{Tr}(S) = d$, resulting in,

$$\mathcal{M}_2 = \frac{1}{d(d+1)}(\mathbb{I} + S). \quad (\text{A.8})$$

Incorporating this result into the fidelity expression, we find

$$F(V, U) = \frac{1}{d(d+1)} \text{Tr} \left[(W \otimes W^\dagger)(\mathbb{I} + S) \right], \quad (\text{A.9})$$

where we can evaluate the two terms using the identities $\text{Tr}(A \otimes B) = \text{Tr}(A)\text{Tr}(B)$ and $\text{Tr}((A \otimes B)S) = \text{Tr}(AB)$, the latter of which is highly relevant to the mechanism of virtual distillation, described in chapter ???. This results in the final expression for the fidelity of a unitary error channel with respect to a target unitary, given above by equation A.3.

A.6 ZZ-type noise

In the main text we explore a separable error channel that induces correlations between qubits in a register. Here we shall explore entangling type noise, which although not prevalent on the ion-trap qubit platform, is a dominant error source in superconducting qubits.

The always on ZZ-type error common to superconducting qubits is described by the Hamiltonian,

$$H_{ZZ} = \frac{J_{ZZ}}{2} Z \otimes Z, \quad (\text{A.10})$$

where J_{ZZ} is the coupling strength. Generally, H_{ZZ} will couple the subrepresentations of the 3-irrep, given in table 1.1. We can show the qualitative difference in the subspace dynamics by plotting the Liouville representation of the twirled *common noise*, and the ZZ-type noise. Specifically we plot $\tilde{\Lambda}^m$, where m is the sequence length. Figure 1.5 plots the maps in the Schur basis ordered so isotypic representations are adjacent to one another. Orange blocks have entries of 1, light green blocks are 0, and red blocks represent some decay parameters dependent on m . Panel (a) shows the fully diagonal *common noise* transfer matrix, where we can see the invariant trace subspace that was noted in the main text. Panel (b) shows the general case for twirled ZZ-type noise, given by equation A.10. The off-diagonal entries exactly couple the three subspaces of the 3-irrep. We note that the resulting block diagonal form is inconvenient to express as a general multi-exponential decay profile as given in the main text for *common* and local depolarising type noise. However, the resulting block diagonal Liouville form may be used directly with experimental data to fit these parameters of interest. Further, we may still explore the expected asymptotic decays due to this model, and find that under dominant ZZ-type noise, we expect an asymptote of $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/2$.

We may recover a simple diagonal Liouville form in the case of stochastic ZZ noise, given by,

$$\mathcal{E}_{ZZ}(\rho) = (1 - p_{ZZ})\rho + p_{ZZ}(Z \otimes Z)\rho(Z \otimes Z), \quad (\text{A.11})$$

where p_{ZZ} is the probability of a correlated phase flip error on both qubits. In the irrep basis, the following sector decays are found for the stochastic ZZ error,

$$\begin{aligned} \alpha_L = \alpha_R = \alpha_{so} = \alpha_{as} &= 1 - 4p_{ZZ}/3, \\ \alpha_{tr} = \alpha_{dt} &= 1. \end{aligned} \quad (\text{A.12})$$

We show this dependence in figure A.1 (c). The same intermediate asymptote of $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 0.5$ is expected.

The difference in correlated decay structure due to *common*, depolarising, and ZZ -type errors suggests that twirling over $Cl_1^{\otimes 2}$ both simplifies a generic error map to be measurable via RB, but also, retains some noise-dependent structure of the correlated decay profiles. Designing optimal initial and final states, as well as optimal sampling strategies for extracting a larger set of error parameters covering multiple physical error mechanisms is left for future work. A possible strategy to avoid introducing arbitrary error channels, is to be agnostic of the physical error mechanisms, and rather directly extract the $\vec{\alpha}$ and $\vec{m}$ parameters described in section 1.3.2. An analogous protocol to this is described by character RB [18].

References

- [1] Jens Eisert et al. “Quantum certification and benchmarking”. en. In: *Nat Rev Phys* 2.7 (2020).
- [2] Easwar Magesan, J. M. Gambetta, and Joseph Emerson. “Scalable and Robust Randomized Benchmarking of Quantum Processes”. In: *Phys. Rev. Lett.* 106.18 (2011).
- [3] Alexander Erhard et al. “Characterizing large-scale quantum computers via cycle benchmarking”. en. In: *Nat Commun* 10.1 (2019).
- [4] Akel Hashim et al. “A Practical Introduction to Benchmarking and Characterization of Quantum Computers”. en. In: *PRX Quantum* 6.3 (2025). arXiv:2408.12064 [quant-ph].
- [5] Mohan Sarovar et al. “Detecting crosstalk errors in quantum information processors”. en. In: *Quantum* 4 (2020). arXiv:1908.09855 [quant-ph].
- [6] Kenneth Rudinger et al. “Experimental Characterization of Crosstalk Errors with Simultaneous Gate Set Tomography”. en. In: *PRX Quantum* 2.4 (2021).
- [7] Robin Harper, Steven T. Flammia, and Joel J. Wallman. “Efficient learning of quantum noise”. In: *Nat. Phys.* 16.12 (2020). arXiv:1907.13022 [quant-ph].
- [8] Robin Harper and Steven T. Flammia. “Learning Correlated Noise in a 39-Qubit Quantum Processor”. en. In: *PRX Quantum* 4.4 (2023).
- [9] Erik Nielsen et al. “Efficient flexible characterization of quantum processors with nested error models”. en. In: *New J. Phys.* 23.9 (2021).
- [10] E. Knill et al. “Randomized benchmarking of quantum gates”. In: *Phys. Rev. A* 77.1 (2008).
- [11] Jay M. Gambetta et al. “Characterization of Addressability by Simultaneous Randomized Benchmarking”. In: *Phys. Rev. Lett.* 109.24 (2012).
- [12] David C. McKay et al. *Correlated Randomized Benchmarking*. arXiv:2003.02354 [quant-ph]. 2020.
- [13] David C. McKay et al. “Benchmarking Quantum Processor Performance at Scale”. In: (2023). _eprint: 2311.05933.
- [14] Joel J. Wallman et al. “Estimating the Coherence of Noise”. en. In: *New J. Phys.* 17.11 (2015). arXiv:1503.07865 [quant-ph].
- [15] Balázs Gulácsi, Joris Kattemölle, and Guido Burkard. “Characterizing correlated noise with single-qubit operations”. In: *Phys. Rev. Res.* 7.4 (2025).
- [16] Ethan R. Clements et al. “Lifetime-Limited Interrogation of Two Independent ^{27}Al Clocks Using Correlation Spectroscopy”. In: *Phys. Rev. Lett.* 125.24 (2020).

- [17] R. T. Sutherland et al. *Subspace Leakage Error Randomized Benchmarking of Mølmer-Sørensen Gates*. en. arXiv:2510.09508 [quant-ph]. 2025.
- [18] Jonas Helsen et al. “A new class of efficient randomized benchmarking protocols”. en. In: *npj Quantum Inf* 5.1 (2019).
- [19] Jonas Helsen, Joel J. Wallman, and Stephanie Wehner. “Representations of the multi-qubit Clifford group”. en. In: *Journal of Mathematical Physics* 59.7 (2018). arXiv:1609.08188 [quant-ph].
- [20] Joel J Wallman and Steven T Flammia. “Randomized benchmarking with confidence”. en. In: *New J. Phys.* 16.10 (2014).
- [21] Peng Zhao et al. “Quantum Crosstalk Analysis for Simultaneous Gate Operations on Superconducting Qubits”. en. In: *PRX Quantum* 3.2 (2022).
- [22] Ángel Rivas and Markus Müller. “Quantifying spatial correlations of general quantum dynamics”. en. In: *New J. Phys.* 17.6 (2015).
- [23] C. A. Sackett et al. “Experimental entanglement of four particles”. en. In: *Nature* 404.6775 (2000).
- [24] D.F.V. James. “Quantum dynamics of cold trapped ions with application to quantum computation”. en. In: *Applied Physics B: Lasers and Optics* 66.2 (1998).